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(54) **SYSTEMS AND METHODS FOR COOKING PIZZA**

231,911 A 9/1880 Holliday
325,645 A 9/1885 Armington
371,892 A 10/1887 Kahn

(Continued)

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FOREIGN PATENT DOCUMENTS

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AU 511593 B2 8/1980
AU 2014208185 B2 8/2015

(Continued)

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OTHER PUBLICATIONS

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See application file for complete search history.

(56) **References Cited**

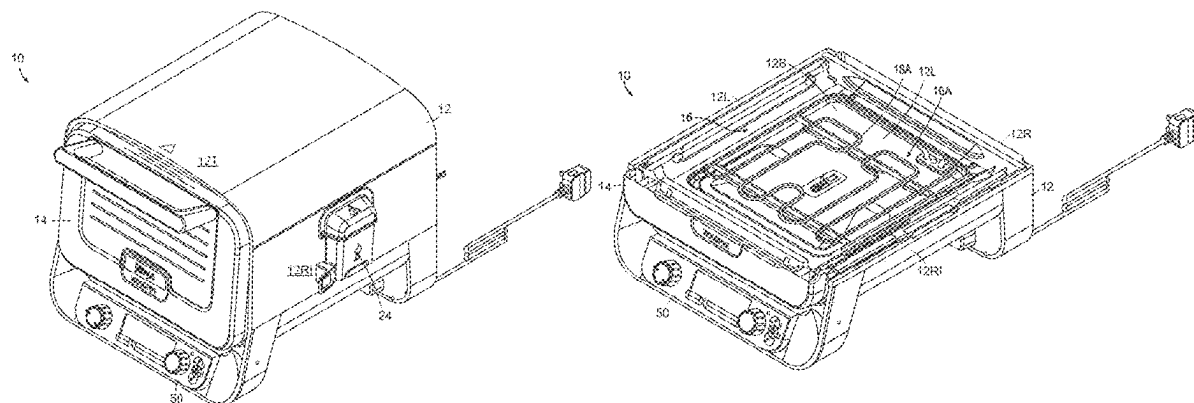
U.S. PATENT DOCUMENTS

195,383 A 9/1877 Montross
213,694 A 3/1879 Sandner, Jr.

(57) **ABSTRACT**

In general, cooking devices having at least one electric heating element and that are configured to cook various foods, including pizza, are provided. In some embodiments, the cooking device can include a housing having a base, a movable door coupled to the base that together define an interior cooking chamber. A cooking surface, such as a cooking stone, can be disposed proximate a heating element within the cooking chamber such that food, such as a pizza, can be placed on top of the cooking stone when inserted into the interior chamber. The heating element can be in operable communication with a controller configured to adjust the amount of heat supplied by the heating element to the interior chamber to optimize the cooking of a food, such as pizza, placed inside the interior chamber.

19 Claims, 10 Drawing Sheets



US 12,408,792 B2

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(56)	References Cited			7,626,142 B2	12/2009	Backus et al.	
	U.S. PATENT DOCUMENTS			7,686,010 B2	3/2010	Gustavsen	
				7,800,023 B2 *	9/2010	Burtea	A21B 1/02
							126/92 AC
416,160 A	12/1889	Galbraith		7,851,727 B2 *	12/2010	Burtea	A21B 2/00
418,721 A	1/1890	Hoogerzeil, Jr.					126/92 AC
436,788 A	9/1890	Jewett		8,020,548 B2	9/2011	Bartmann et al.	
456,000 A	7/1891	Graves		8,047,198 B2	11/2011	Meyer et al.	
467,292 A	1/1892	Hoogerzeil		8,047,199 B2	11/2011	Le et al.	
519,606 A	5/1894	Kaempfen, Jr.		8,201,552 B2	6/2012	Ploof et al.	
688,639 A	12/1901	Harton		8,267,257 B2	9/2012	Doyal	
969,340 A	9/1910	Clark		8,308,250 B2	11/2012	Koehler	
1,262,659 A	4/1918	Foster		8,327,837 B2	12/2012	Nam et al.	
1,429,870 A	9/1922	Arvin		D678,005 S	3/2013	Zemel et al.	
1,507,778 A	9/1924	Keavey et al.		8,439,028 B2	5/2013	Haberkamm et al.	
1,582,543 A	4/1926	Rowden		8,481,894 B2	7/2013	Robin et al.	
1,627,405 A	5/1927	Prenveille et al.		8,490,801 B2	7/2013	Smith et al.	
1,628,895 A	5/1927	McKenna		8,573,720 B2	11/2013	Wittgrebe	
1,632,719 A	6/1927	White		8,578,927 B2	11/2013	Gustavsen	
1,776,475 A	9/1930	Ernest		8,602,019 B2	12/2013	Baker et al.	
1,785,514 A	12/1930	Chandler et al.		8,646,444 B2	2/2014	Williams et al.	
1,793,955 A	2/1931	Otte		8,669,500 B2 *	3/2014	Hensel	A47J 37/0807
1,831,652 A	11/1931	Chandler et al.					219/483
1,851,854 A	3/1932	Lindemann et al.		8,733,862 B1	5/2014	Armstrong et al.	
1,894,268 A	1/1933	Forsyth et al.		8,820,314 B1	9/2014	Johnson et al.	
1,938,470 A	12/1933	Jacob et al.		8,820,689 B2	9/2014	Reidt et al.	
1,958,348 A	5/1934	Odis		8,899,426 B2	12/2014	Dunn	
1,970,343 A	8/1934	Sherman		8,919,338 B2	12/2014	McNamee et al.	
2,132,737 A	10/1938	Kahn		8,919,339 B2	12/2014	Mazzetti et al.	
2,155,654 A	4/1939	Haley		9,022,018 B2	5/2015	Hensel	
2,255,129 A	9/1941	Rogers		9,052,118 B2	6/2015	Metcalfe et al.	
2,296,950 A	9/1942	Roedl		9,078,517 B2	7/2015	Rehage	
2,352,613 A	7/1944	Bradbury		9,089,005 B2 *	7/2015	Boedicker	H05B 1/0263
2,447,184 A	8/1948	Jones		9,175,860 B2	11/2015	Padilha et al.	
2,486,564 A	11/1949	Kamin		9,468,336 B2	10/2016	McQuillan	
2,657,110 A	10/1953	Feder		9,541,295 B2	1/2017	Adelmann et al.	
2,682,263 A	6/1954	Brodbeck		9,631,820 B2	4/2017	Hensel	
2,744,994 A	5/1956	Harry et al.		9,681,773 B2	6/2017	McKee et al.	
3,016,276 A	1/1962	Morse		9,844,297 B2	12/2017	Volatier et al.	
3,106,202 A	10/1963	Arduna		9,863,645 B2	1/2018	Adelmann et al.	
3,146,338 A	8/1964	Jerome		9,874,353 B2	1/2018	Thabit	
3,781,525 A	12/1973	Joeckel		9,907,435 B2	3/2018	Kohler et al.	
3,830,220 A	8/1974	Demetreion		9,939,160 B2	4/2018	Kim et al.	
4,149,518 A	4/1979	Schmidt et al.		9,964,312 B2	5/2018	Deng et al.	
4,637,373 A	1/1987	Shirai et al.		10,101,038 B2	10/2018	Kim et al.	
4,649,810 A	3/1987	Wong		10,132,502 B2	11/2018	Contarino, Jr.	
4,683,871 A	8/1987	Salvi		10,247,424 B2	4/2019	Adelmann et al.	
4,829,158 A	5/1989	Burnham		10,260,757 B2	4/2019	Stewart et al.	
4,865,219 A	9/1989	Logan et al.		10,352,073 B2	7/2019	Lee et al.	
4,892,085 A	1/1990	Salvi		10,383,477 B2	8/2019	Payen et al.	
D337,032 S	7/1993	Galati		10,405,699 B2	9/2019	Contarino, Jr.	
5,338,922 A	8/1994	Kim et al.		10,429,077 B2	10/2019	Kaiser et al.	
5,355,779 A	10/1994	O'Brien et al.		10,520,202 B2	12/2019	Delrue et al.	
5,365,918 A	11/1994	Smith et al.		10,524,616 B2	1/2020	Contarino, Jr.	
5,434,390 A	7/1995	McKee et al.		10,537,199 B2	1/2020	Knappenberger et al.	
5,522,308 A *	6/1996	Kayashima	G05D 23/2401	10,548,396 B1	2/2020	Hong	
			219/442	10,551,071 B2	2/2020	Bianchi et al.	
5,674,421 A	10/1997	Beaver et al.		10,561,277 B1 *	2/2020	Swayne	A23L 5/17
6,140,621 A	10/2000	Ho et al.		10,575,680 B2	3/2020	Fagg et al.	
RE37,152 E	5/2001	O'Brien et al.		10,591,168 B2	3/2020	Yang et al.	
6,437,291 B1	8/2002	Hopponen		10,634,362 B2	4/2020	Trice et al.	
6,444,954 B1	9/2002	Blankenship		10,641,500 B2	5/2020	Yun	
6,509,550 B1	1/2003	Li		10,655,864 B2	5/2020	Howard et al.	
6,621,053 B1	9/2003	Wensink et al.		10,682,010 B2	6/2020	Hoare et al.	
6,734,403 B2	5/2004	Baker et al.		10,712,018 B2	7/2020	Adelmann et al.	
6,930,286 B2	8/2005	Kim		10,718,671 B2	7/2020	Allen	
7,026,579 B2 *	4/2006	Burtea	A21B 1/48	D892,532 S	8/2020	Gushwa et al.	
			99/443 C	10,765,119 B2	9/2020	Johnson et al.	
7,087,862 B1	8/2006	Shaffer et al.		10,808,944 B2	10/2020	Bianchi et al.	
7,105,778 B1	9/2006	DeLong et al.		10,837,652 B2	11/2020	Balekundri et al.	
7,173,217 B2	2/2007	De'Longhi		10,842,315 B1	11/2020	Swinney	
7,213,592 B2	5/2007	Fischhaber et al.		10,845,062 B2	11/2020	Harvard et al.	
7,225,730 B2	6/2007	Backus et al.		10,905,281 B2	2/2021	Delrue et al.	
7,282,672 B2	10/2007	Wurm		10,941,945 B2	3/2021	Tapaninaho	
D563,179 S	3/2008	Minidis		11,026,542 B2	6/2021	Pruitt et al.	
D564,283 S	3/2008	Leung et al.		11,035,576 B2	6/2021	Johnson et al.	
7,424,849 B2	9/2008	Backus et al.		11,045,047 B2	6/2021	Popeil et al.	
7,448,508 B2	11/2008	Babucke et al.		11,071,403 B2	7/2021	Staton et al.	
D585,235 S	1/2009	Tetreault et al.		D928,549 S	8/2021	Yamamoto	

(56)

References Cited

U.S. PATENT DOCUMENTS

11,076,717 B2	8/2021	Glucksman	2020/0345176 A1	11/2020	Bert
11,105,515 B2	8/2021	Lee et al.	2020/0405085 A1	12/2020	Dos Santos et al.
11,112,125 B2	9/2021	Stewart et al.	2020/0405086 A1	12/2020	Dos Santos et al.
11,129,244 B2	9/2021	Hoare et al.	2021/0015300 A1	1/2021	Nernberger et al.
11,172,780 B2	11/2021	Home	2021/0116134 A1	4/2021	Hu et al.
11,190,015 B2	11/2021	Marsh-Croft et al.	2021/0121004 A1*	4/2021	Elliston A47J 27/0804
11,199,331 B2	12/2021	Mills et al.	2021/0153508 A1	5/2021	Gozney
11,213,165 B2	1/2022	Kammerer	2021/0161151 A1	6/2021	Thorogood et al.
11,286,698 B2	3/2022	Braden et al.	2021/0204371 A1	7/2021	Asami
11,304,421 B2	4/2022	Handwerker	2021/0222888 A1	7/2021	Ceron et al.
11,311,143 B2	4/2022	Peter	2021/0274968 A1	9/2021	Steiner et al.
11,452,404 B2	9/2022	Anderson	2021/0293416 A1	9/2021	Homburg et al.
11,454,677 B2	9/2022	Knappenberger et al.	2021/0369051 A1	12/2021	Thorogood et al.
11,464,360 B2	10/2022	Goldberg	2021/0381695 A1	12/2021	Rehage et al.
11,478,107 B2	10/2022	Volatier	2021/0388992 A1	12/2021	Thorogood et al.
11,523,616 B2	12/2022	Mecsaci	2021/0388993 A1	12/2021	Ceron et al.
11,536,048 B2	12/2022	Kim et al.	2021/0392904 A1	12/2021	Raio et al.
11,598,532 B2	3/2023	Reid et al.	2022/0007689 A1	1/2022	Baldwin
D983,602 S	4/2023	Pruitt et al.	2022/0015575 A1	1/2022	Harper
11,647,861 B2*	5/2023	Elliston A47J 27/0802	2022/0087473 A1	3/2022	Kim et al.
		99/331	2022/0090794 A1	3/2022	Yun et al.
11,674,691 B2*	6/2023	Polster H05B 1/0263	2022/0095834 A1	3/2022	Thorogood et al.
		219/391	2022/0120445 A1	4/2022	Murad
11,732,895 B1	8/2023	Chow et al.	2022/0160172 A1	5/2022	Kwang et al.
11,753,856 B2	9/2023	Shin et al.	2022/0167789 A1	6/2022	Kim et al.
2003/0015518 A1*	1/2003	Baker F24C 7/087	2022/0192421 A1	6/2022	Bert
		219/486	2022/0223025 A1	7/2022	Allen, Sr.
2003/0056658 A1*	3/2003	Jones A47J 37/0857	2022/0304505 A1	9/2022	Tapaninaho
		99/443 C	2022/0322874 A1	10/2022	Marsh-Croft et al.
2005/0217658 A1	10/2005	Sozo et al.	2022/0322877 A1	10/2022	Greer et al.
2005/0274373 A1*	12/2005	Pesonen F27B 9/2407	2022/0361711 A1	11/2022	Huang et al.
		126/21 A	2023/0000284 A1	1/2023	Smith et al.
2006/0027559 A1*	2/2006	Kim F24C 15/325	2023/0038171 A1	2/2023	Stidston et al.
		219/400	2023/0309745 A1	10/2023	Martin et al.
2007/0277678 A1	12/2007	Mangano			
2010/0006558 A1	1/2010	McLoughlin et al.			
2010/0176123 A1*	7/2010	Mihara H05B 6/705			
		219/746			
2011/0283895 A1*	11/2011	Veltrop A47J 39/006			
		392/416			
2012/0192851 A1	8/2012	Hensel et al.			
2013/0167829 A1	7/2013	Lin			
2013/0167830 A1	7/2013	Lin			
2013/0233182 A1	9/2013	Hunt et al.			
2014/0021191 A1	1/2014	Moon et al.			
2014/0035452 A1	2/2014	Nickelson et al.			
2014/0196609 A1	7/2014	Snyman			
2014/0220196 A1*	8/2014	Veloo A23L 5/10			
		426/233			
2015/0292749 A1*	10/2015	Kim F24C 7/085			
		99/335			
2016/0003294 A1	1/2016	Lindinger			
2016/0097542 A1*	4/2016	Armstrong F24C 7/082			
		99/337			
2017/0238751 A1	8/2017	Vengroff			
2017/0303685 A1	10/2017	Rehage et al.			
2018/0020679 A1*	1/2018	Froelicher A21B 1/40			
		219/392			
2018/0317705 A1	11/2018	Roy			
2018/0333004 A1	11/2018	De' Longhi et al.			
2019/0056117 A1	2/2019	Rehage et al.			
2019/0059646 A1	2/2019	Uno et al.			
2019/0107291 A1	4/2019	Cetintas			
2019/0186756 A1	6/2019	Tiefnig et al.			
2019/0290062 A1	9/2019	Prieto et al.			
2019/0316781 A1	10/2019	Kim et al.			
2019/0327796 A1	10/2019	Hannah et al.			
2020/0201414 A1	6/2020	Knappenberger et al.			
2020/0224882 A1	7/2020	Kim et al.			
2020/0224883 A1	7/2020	Kim et al.			
2020/0224884 A1	7/2020	Kwang et al.			
2020/0229276 A1	7/2020	Kim et al.			
2020/0284438 A1	9/2020	Gigante et al.			
2020/0315389 A1	10/2020	Stewart et al.			

FOREIGN PATENT DOCUMENTS

AU	2020202899 A1	11/2020
AU	2020206084 B9	5/2023
CA	2552839 A1	1/2007
CA	2867330 A1	10/2013
CA	2922423 A1	10/2015
CN	102112811 A	6/2011
CN	102362123 A	2/2012
CN	101953645 B	1/2016
CN	104172955 B	1/2017
CN	206053603 U	3/2017
CN	106821025 A	6/2017
CN	106821031 A	6/2017
CN	104153157 B	8/2017
CN	107374390 A	11/2017
CN	107647794 A	2/2018
CN	107684357 A	2/2018
CN	107692865 A	2/2018
CN	107788860 A	3/2018
CN	207355968 U	5/2018
CN	108309073 A	7/2018
CN	109691902 A	4/2019
CN	208973522 U	6/2019
CN	107514662 B	9/2019
CN	110412917 A	11/2019
CN	110513004 A	11/2019
CN	110693348 A	1/2020
CN	110811351 A	2/2020
CN	110840284 A	2/2020
CN	110953616 A	4/2020
CN	110973995 A	4/2020
CN	111012199 A	4/2020
CN	210471883 U	5/2020
CN	108552970 B	7/2020
CN	110123157 B	8/2020
CN	112244658 A	1/2021
CN	112690666 A	4/2021
CN	110403452 B	5/2021
CN	110664268 B	5/2021
CN	110805936 B	5/2021
CN	112806866 A	5/2021
CN	112971546 A	6/2021
CN	112971550 A	6/2021

(56)

References Cited

FOREIGN PATENT DOCUMENTS

CN	113143047	A	7/2021
CN	112263140	B	10/2021
CN	113944942	A	1/2022
CN	111010856	B	3/2022
CN	114224197	A	3/2022
DE	2850366	A1	5/1980
DE	102010008420	A1	8/2011
EP	0147815	A2	7/1985
EP	0151192	A1	8/1985
EP	0443329	A2	8/1991
EP	0578360	B1	12/1995
EP	1400195	B1	2/2005
EP	1403587	B1	5/2006
EP	1466546	B1	11/2006
EP	1580145	B1	4/2007
EP	1419722	B1	9/2007
EP	1800579	B1	12/2008
EP	1865264	B1	5/2009
EP	1795096	B1	6/2009
EP	1776914	B1	9/2009
EP	2042811	B1	3/2011
EP	2096359	B1	5/2011
EP	2204115	B1	8/2011
EP	2093497	B1	5/2013
EP	2703736	B1	10/2015
EP	2093493	B1	2/2016
EP	2093495	B1	2/2016
EP	1553350	B1	3/2016
EP	2538144	B1	8/2016
EP	1840473	B1	11/2016
EP	2615380	B1	3/2017
EP	2843316	B1	6/2017
EP	2950002	B1	7/2017
EP	1918644	B1	10/2017
EP	2757322	B1	12/2017
EP	2292980	B1	3/2018
EP	2339240	B1	8/2018
EP	2739191	B1	9/2018
EP	3375288	A1	9/2018
EP	3031328	B1	10/2018
EP	3399836	A1	11/2018
EP	1553347	B1	1/2019
EP	2913593	B1	1/2019
EP	2730848	B1	6/2019
EP	3287050	B1	6/2019
EP	2436990	B1	7/2019
EP	3196856	B1	8/2019
EP	2597378	B1	1/2020
EP	3217760	B1	6/2020
EP	3712508	A1	9/2020
EP	3427616	B1	1/2021
EP	3771867	A1	2/2021
EP	3838067	A1	6/2021
EP	3875858	A1	9/2021
EP	3944794	A1	2/2022
EP	3450855	B1	6/2022
EP	3910242	A4	11/2022
EP	3949815	B1	12/2022

FR	2810067	B3	4/2002
FR	2893832	B1	2/2012
FR	3079600	B1	9/2020
FR	3117325	A1	6/2022
GB	1575264	A	9/1980
GB	2602928	B	11/2022
JP	H05172342	A	7/1993
JP	2008513724	A	5/2008
JP	2019098149	A	6/2019
KR	20200087653	A	7/2020
KR	20200087684	A	7/2020
KR	20200087687	A	7/2020
KR	20210007294	A	1/2021
KR	102213630	B1	2/2021
KR	102364484	B1	2/2022
KR	102368211	B1	2/2022
KR	102369772	B1	3/2022
KR	102432825	B1	8/2022
KR	102522669	B1	4/2023
WO	WO-2006098646	A1 *	9/2006
WO	2007077163	A1	7/2007
WO	2009107016	A1	9/2009
WO	2011120698	A1	10/2011
WO	2012117274	A1	9/2012
WO	2014082828	A1	6/2014
WO	2017013331	A1	1/2017
WO	2017015270	A1	1/2017
WO	2017085393	A1	5/2017
WO	2017087329	A1	5/2017
WO	2018024558	A1	2/2018
WO	2019203444	A1	10/2019
WO	2020043427	A1	3/2020
WO	2020135986	A1	7/2020
WO	2020145756	A1	7/2020
WO	2020145758	A1	7/2020
WO	2020145760	A1	7/2020
WO	2020145764	A1	7/2020
WO	2020256192	A1	12/2020
WO	2021165198	A1	8/2021
WO	2021165200	A1	8/2021
WO	2021165261	A1	8/2021
WO	2021165706	A3	10/2021
WO	2022017792	A1	1/2022
WO	2022051800	A1	3/2022
WO	2022053226	A1	3/2022
WO	2022106474	A1	5/2022
WO	2022216347	A1	10/2022
WO	2023272336	A1	1/2023
WO	2023004464	A1	2/2023

..... A47J 27/026

OTHER PUBLICATIONS

U.S. Appl. No. 18/079,781 entitled “Grill Systems” filed May 16, 2022, 53 pages.

U.S. Appl. No. 18/307,595, filed Apr. 26, 2023, Systems and Methods for Cooking Pizza, Jan. 7, 2025.

U.S. Appl. No. 18/307,583, filed Apr. 26, 2023, Systems and Methods for Cooking Pizza, Jan. 7, 2025.

* cited by examiner

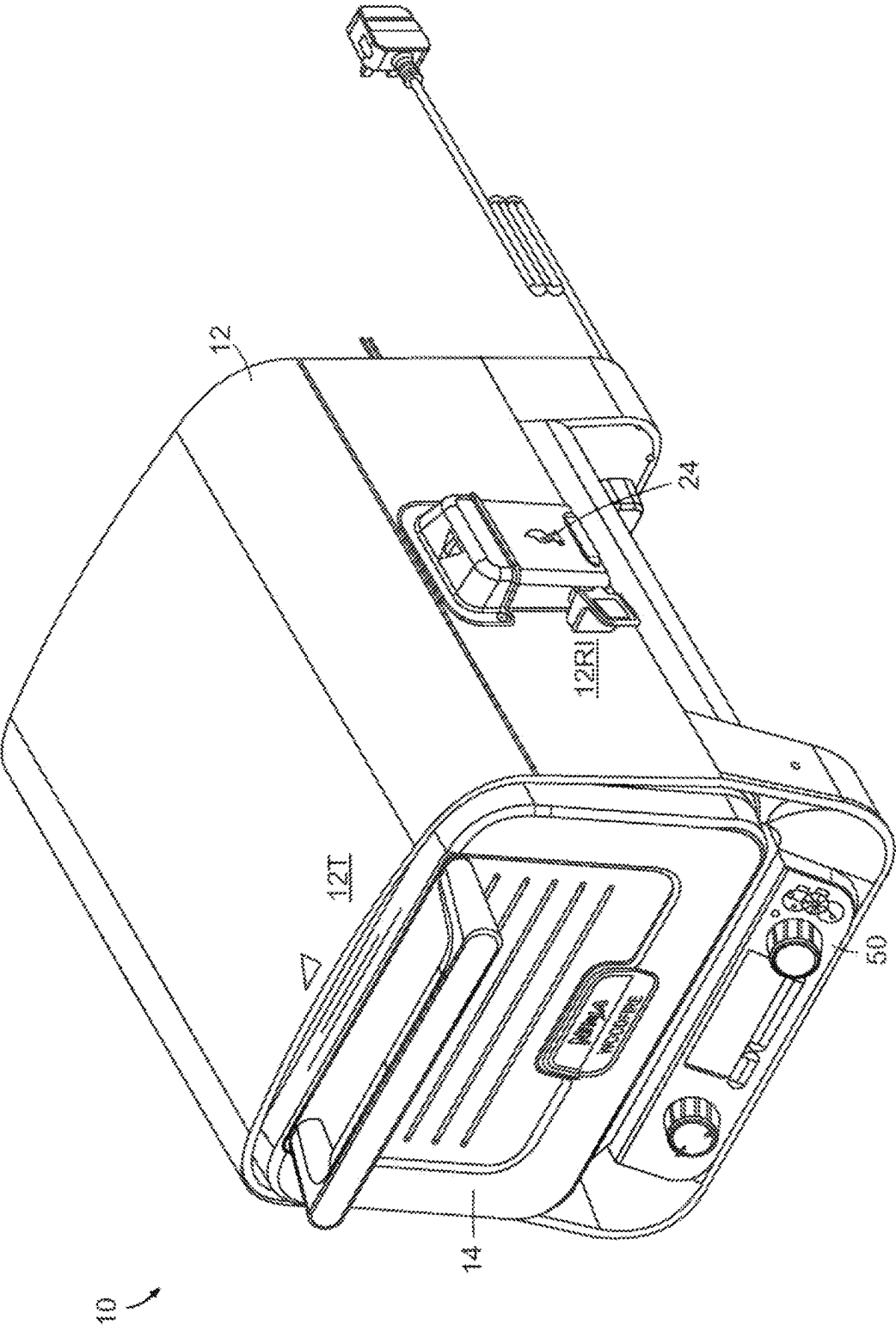


FIG. 1A

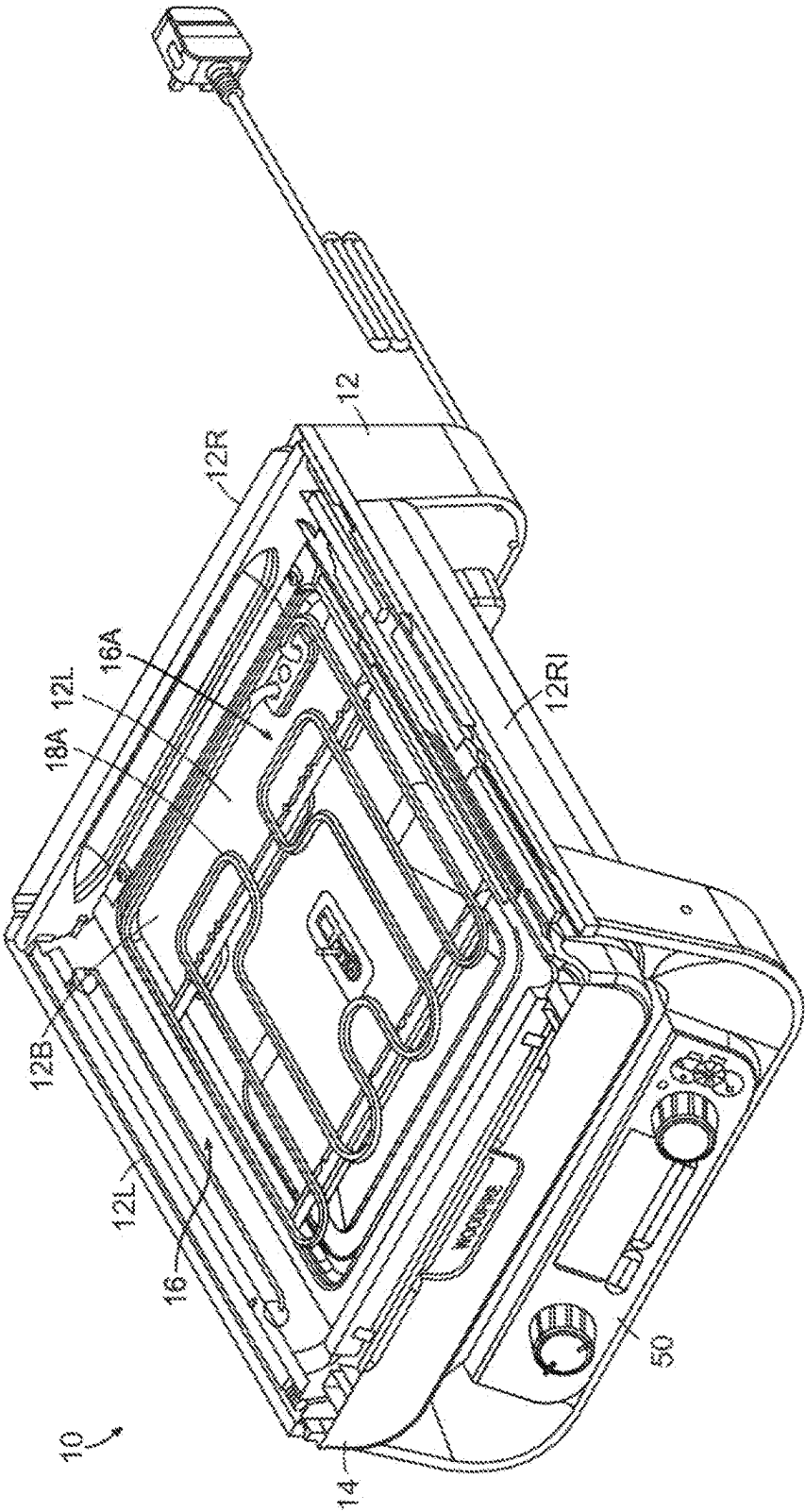


FIG. 1B

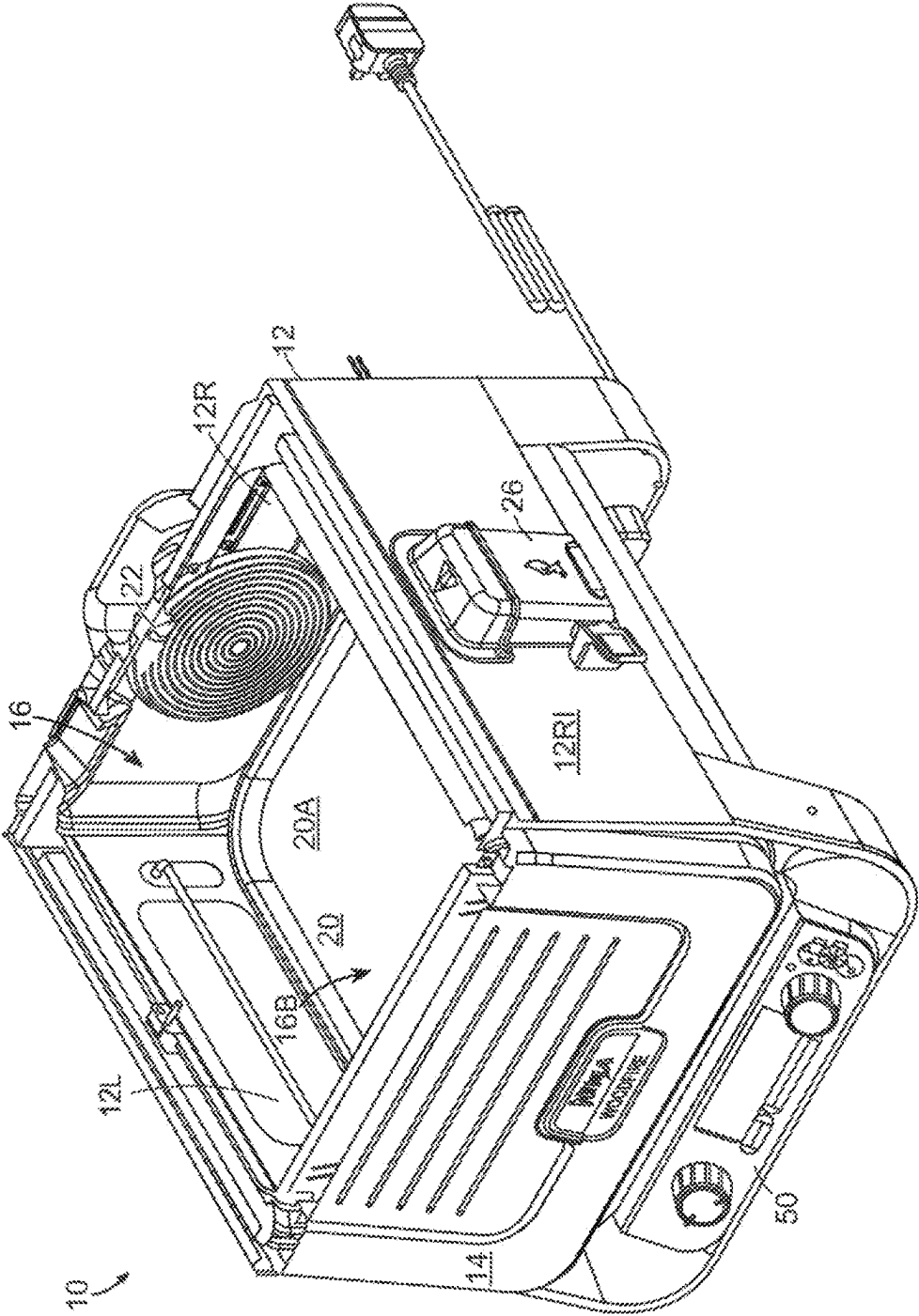


FIG. 1C

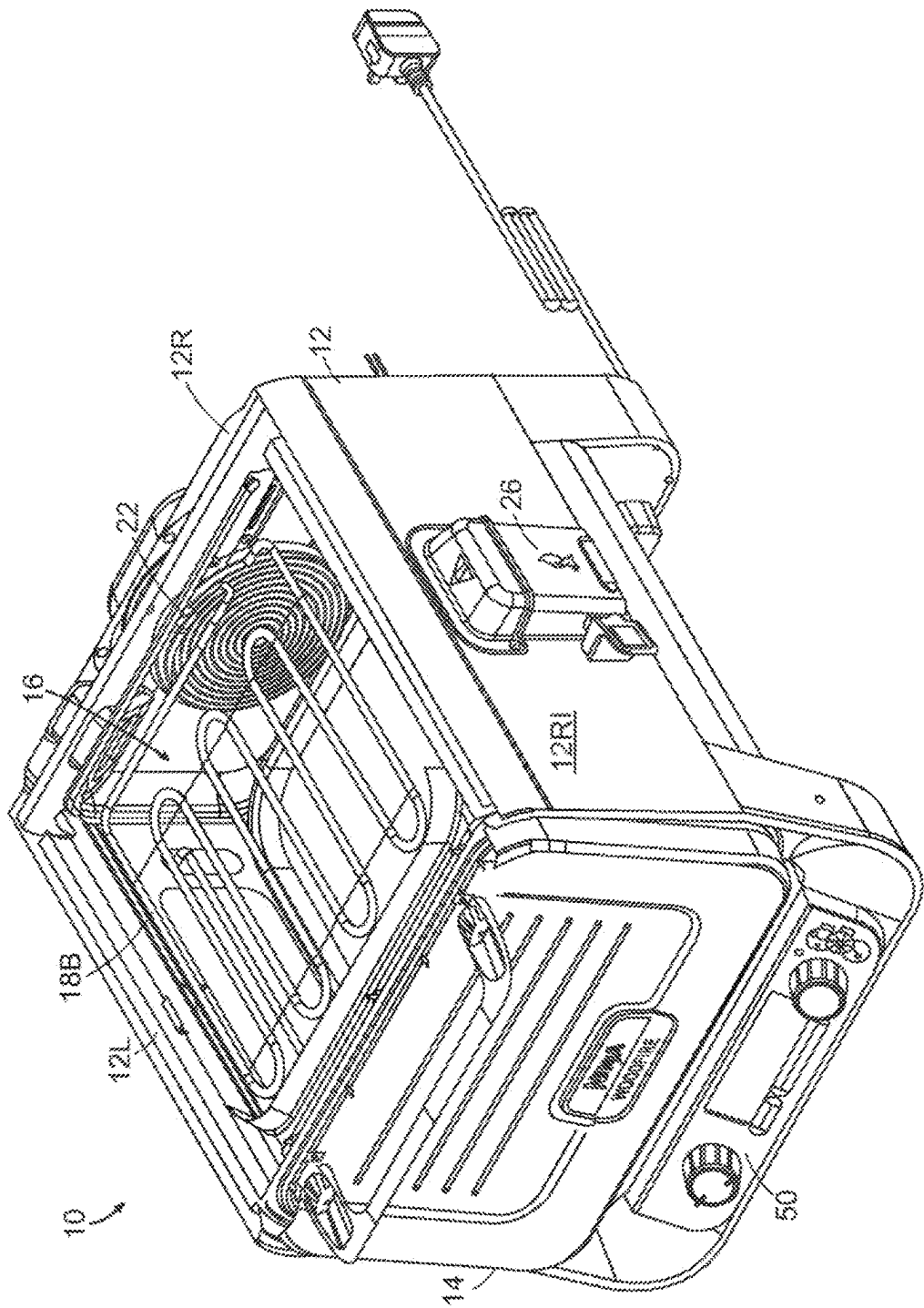


FIG. 1D

50

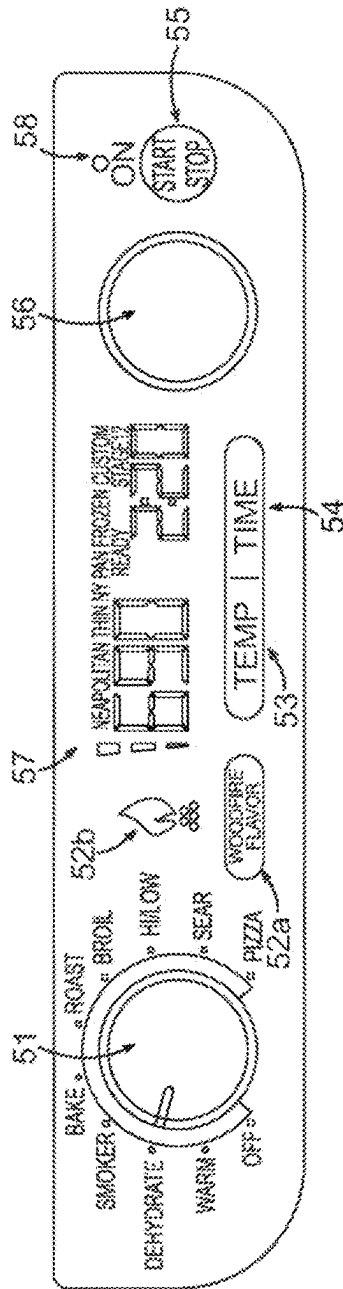


FIG. 1E

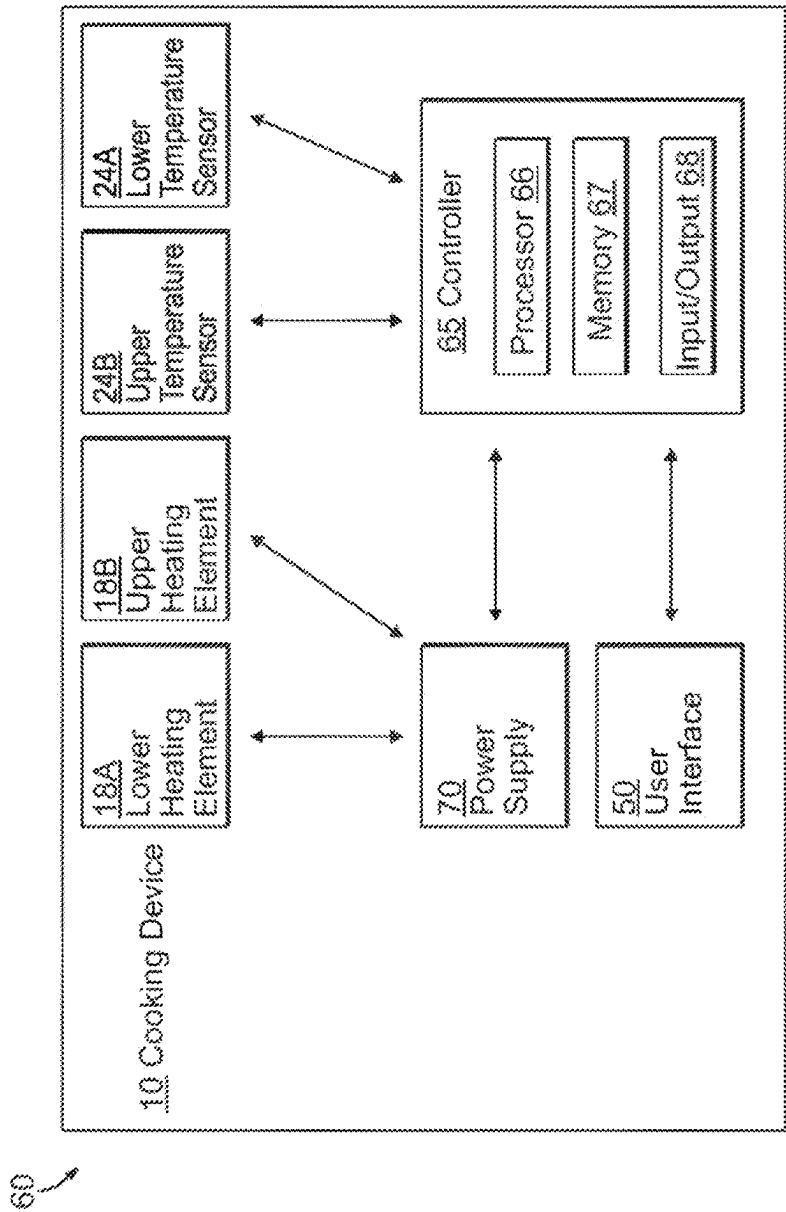
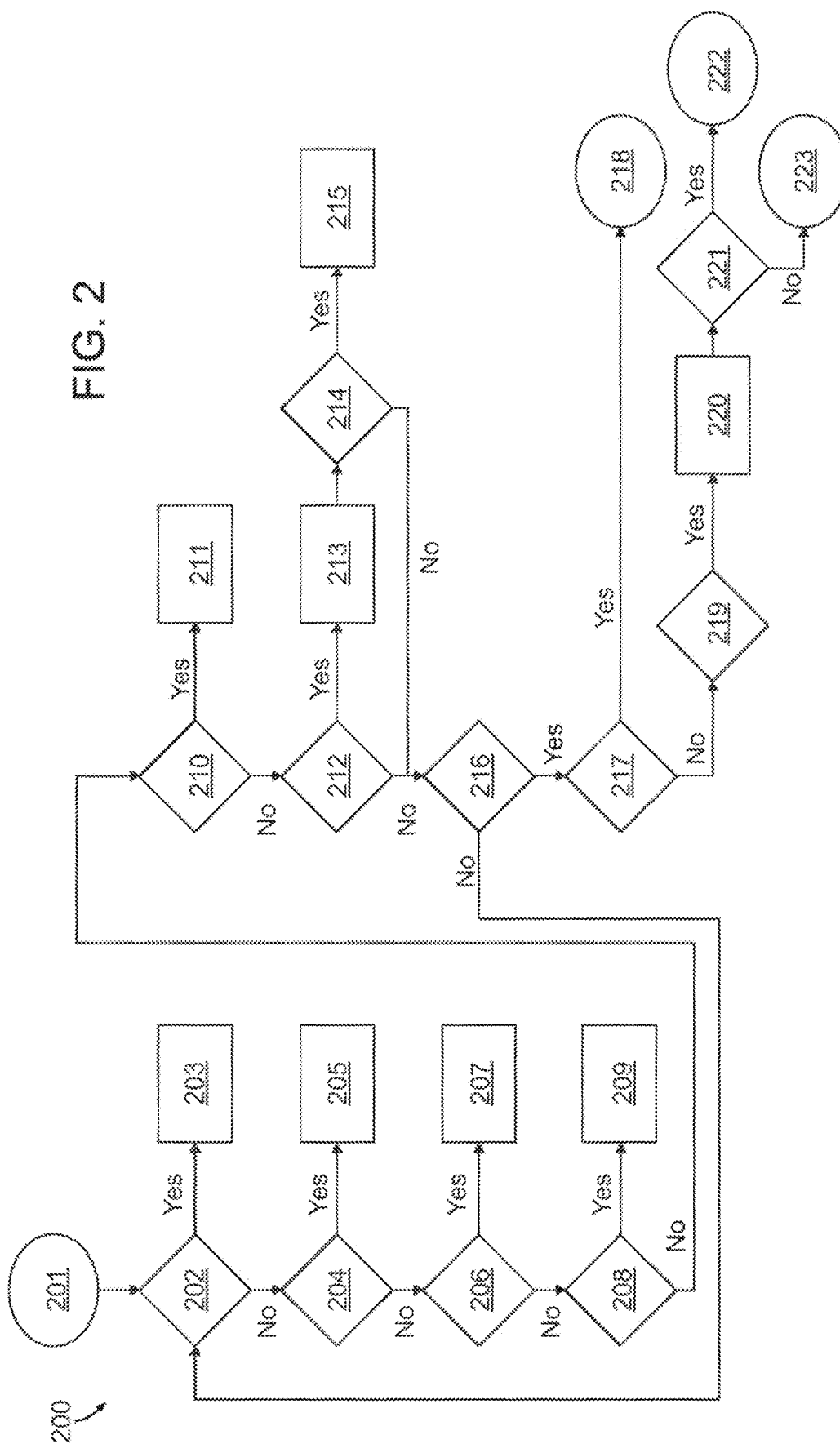


FIG. 1F

FIG. 2



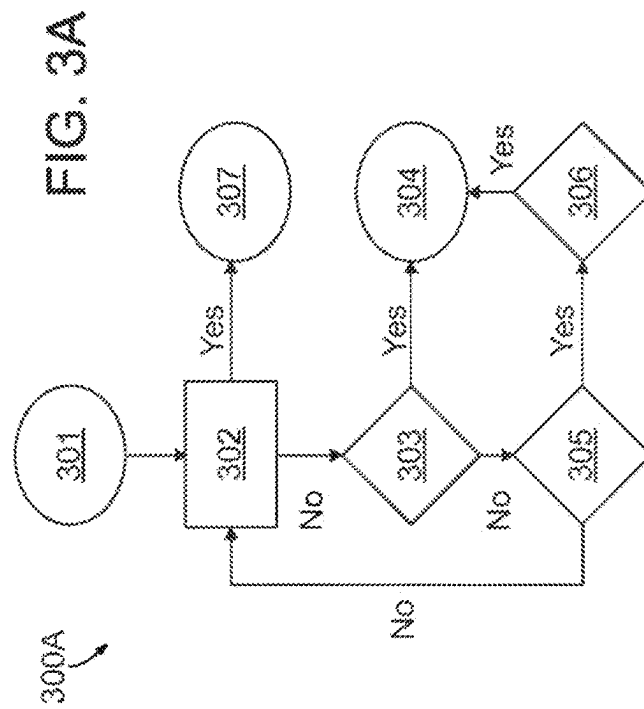
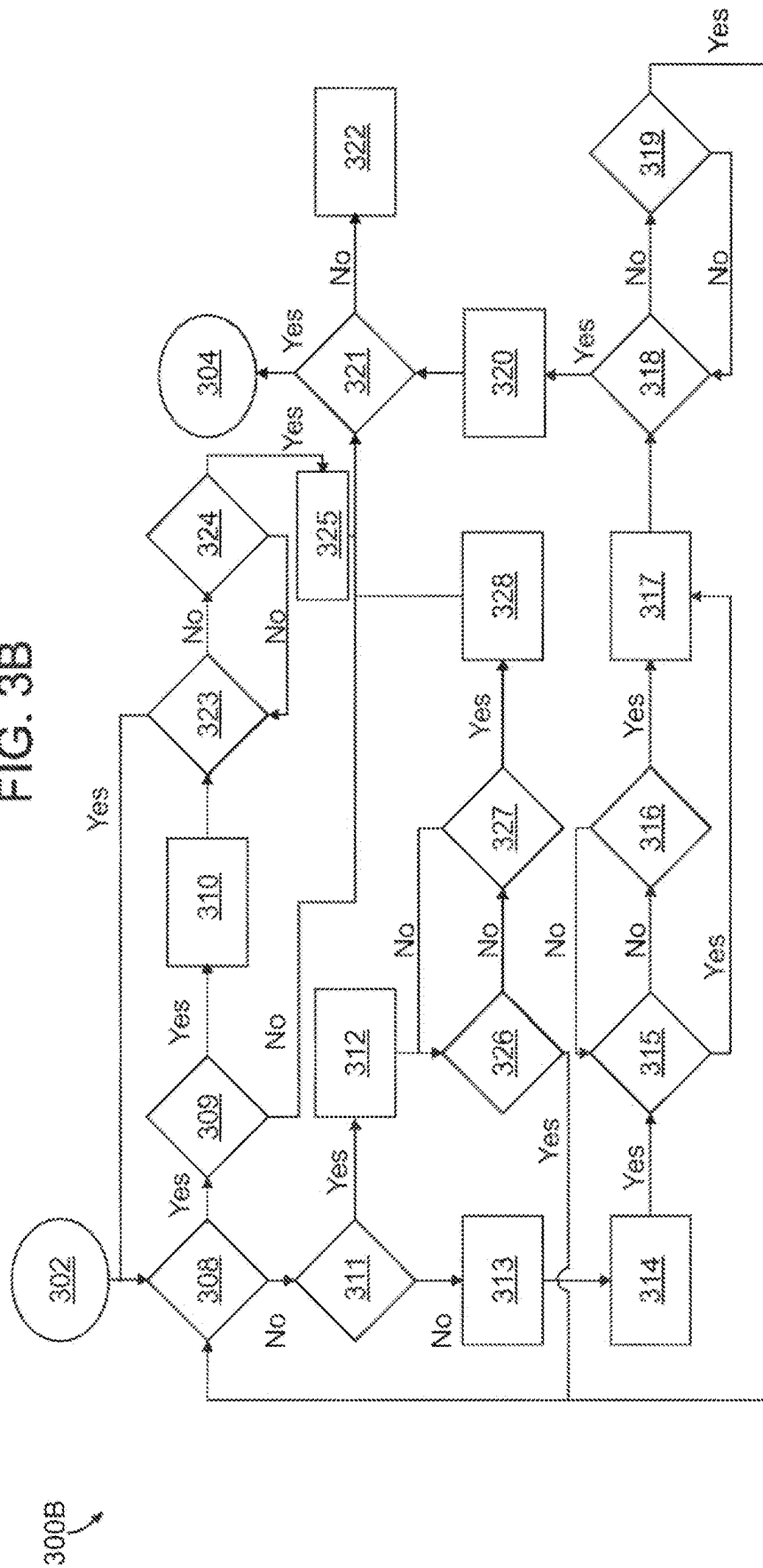


FIG. 3B



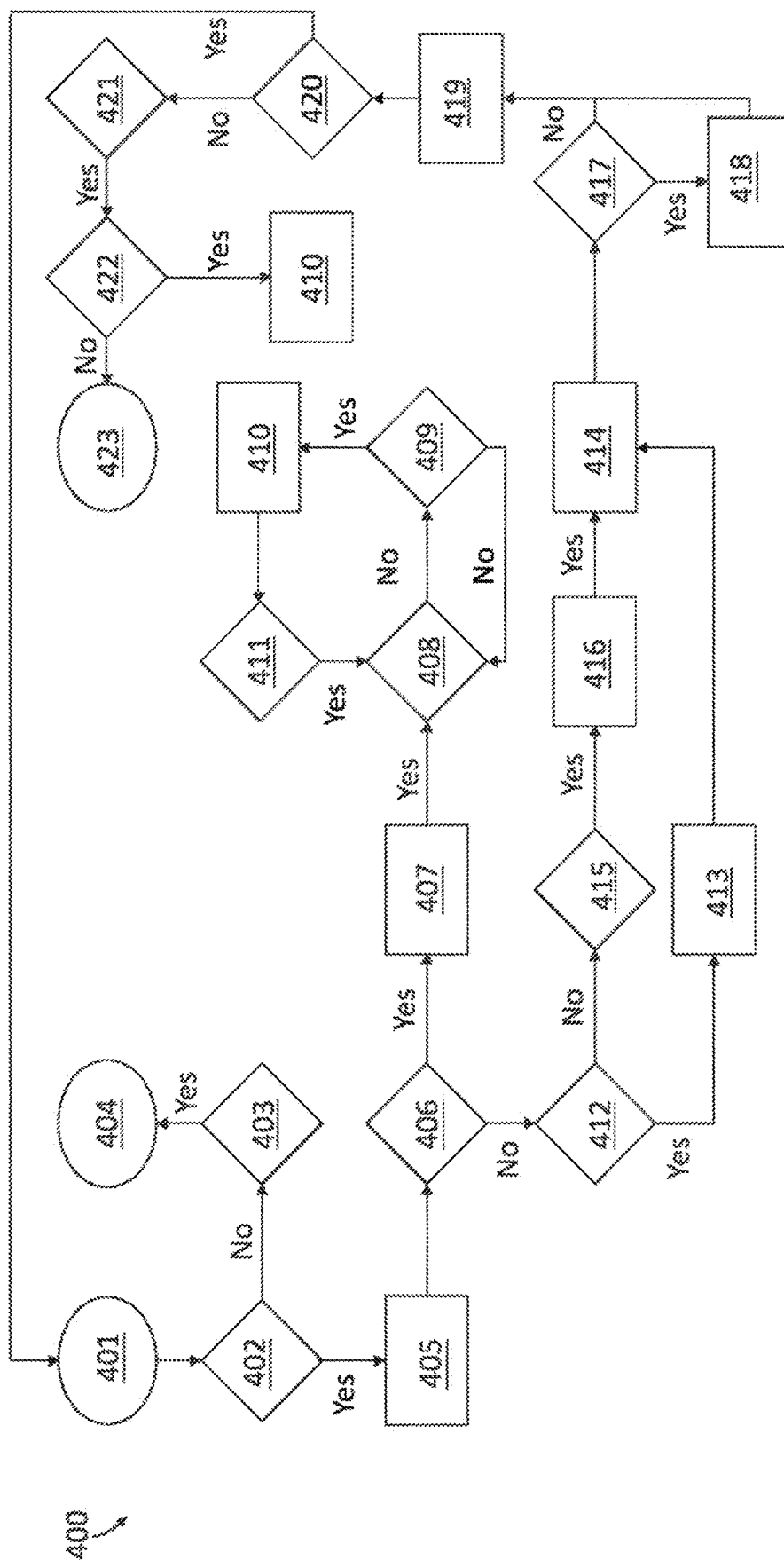


FIG. 4

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SYSTEMS AND METHODS FOR COOKING PIZZA

CROSS REFERENCE TO RELATED APPLICATION

This application is a divisional of U.S. patent application Ser. No. 18/307,595, filed Apr. 26, 2023, and entitled “SYSTEMS AND METHODS FOR COOKING PIZZA”, which is incorporated by reference herein in its entirety.

FIELD

An oven system for cooking food, such as pizza, and methods for using the same, are provided.

BACKGROUND

Pizzas can be made in a variety of styles (e.g., Neapolitan, New York Style, etc.) and can be prepared and cooked using techniques that vary based on the desired type of pizza style. Pizza ovens can provide for the cooking of pizza by using one or more heating elements to transfer heat to the pizza. However, some pizza ovens, such as those that use electric heating elements, may be limited in the ways by which they can transfer heat to the pizza during cooking and thus may be unable to cook a pizza in accordance with the cooking technique required to achieve certain pizza styles. For example, to successfully cook Neapolitan pizzas, the pizza oven must be capable of imparting a high level of heat to the crust of the pizza. Given the high levels of heat needed to cook such a pizza, it can be challenging to repeatedly cook Neapolitan pizzas with a short cycle time.

Accordingly, there remains a need for an improved oven system for cooking various foods, including pizza.

SUMMARY

An oven system for cooking food, such as pizza, and methods for using the same, are provided. Related apparatuses and techniques are also described.

In one aspect, a cooking device is described. The cooking device can include a housing defining an interior chamber, an upper heating element disposed within the interior chamber and on an upper surface of the housing and configured to supply heat to the interior chamber. The cooking device can also include a lower heating element disposed within the interior chamber, on a lower surface of the housing, proximate a cooking surface disposed within the interior chamber and configured to receive food to be cooked, and the lower heating element can be configured to supply heat to the interior chamber. The lower surface can be opposite the upper surface. The cooking device can also include a controller in operable communication with the upper heating element and the lower heating element. The controller can be configured to receive temperature data characterizing a temperature of air within an upper region of the interior chamber above a cooking surface and a temperature of air within a lower region of the interior chamber below the cooking surface, to determine a ratio between an amount of heat to be supplied to the interior chamber by the upper heating element and an amount of heat to be supplied to the interior chamber by the lower heating element based on the temperature data, and to cause the heat to be supplied to the interior chamber by at least one of the upper and lower heating elements based on the determined ratio.

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One or more of the following features can be included in any feasible combination. For example, the controller can be configured to receive data characterizing a mode of cooking the food and to determine the ratio based on the received data. For example, the cooking device can further include an upper temperature sensor disposed within the interior chamber and proximate the upper heating element, and the upper temperature sensor can be configured to measure the temperature of the air within the interior chamber above the cooking surface. For example, the cooking device can further include a lower temperature sensor disposed within the interior chamber and proximate the lower heating element and the cooking surface, and the lower temperature sensor can be configured to measure the temperature of at least one of the cooking surface and the air within the lower region of the interior chamber. For example, the controller can be configured to cause the heat to be supplied to the interior chamber by at least one of the upper and lower heating elements based on the determined ratio when the controller is operating in a pre-heating state. For example, the controller can be configured to adjust the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element after a period of time, and the period of time can be determined based on the determined ratio. For example, the controller can be configured to adjust the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element when the temperature of the air within the lower region of the interior chamber is less than a target temperature.

In another aspect, a system is provided and can include at least one data processor and memory storing instructions configured to cause the at least one data processor to perform operations described herein. The operations can include receiving temperature data characterizing a temperature of air within an upper region of an interior chamber of a cooking device, the upper region above a cooking surface, and characterizing a temperature of air within a lower region of the interior chamber, the lower region below the cooking surface; determining, based on the received temperature data, a ratio between an amount of heat to be supplied to the interior chamber by an upper heating element and an amount of heat to be supplied to the interior chamber by a lower heating element, the upper heating element and the lower heating element being disposed in the interior chamber at opposing surfaces of the interior chamber; and causing heat to be supplied to the interior chamber by at least one of the upper and lower heating elements based on the determined ratio.

One or more of the following features can be included in any feasible combination. For example, the operations can further include receiving mode data characterizing a mode of cooking the food, and wherein the ratio is determined based on the received mode data. For example, the causing of the heat to be supplied to the interior chamber can occur when the cooking device is operating in a pre-heating state. For example, the operations can further include determining a period of time based on the determined ratio, the period of time beginning after the causing of the heat to be supplied to the interior chamber; and adjusting the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element after the determined period of time has elapsed. For example, the operations can further include adjusting the amount of the heat supplied to

the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element when the temperature of the air within the lower region of the interior chamber characterized by the temperature data is less than a target temperature. For example, the operations can further include receiving second temperature data characterizing a second temperature of the air within the interior chamber; determining whether the second temperature characterized by the received second temperature data is greater than or equal to the target temperature; and, in response to determining that the temperature characterized by the received second temperature data is greater than or equal to the target temperature, causing the upper heating element and the lower heating element to deactivate. For example, the operations can further include determining a second period of time based on the determined ratio, the second period of time different from the period of time and beginning after the causing of the heat to be supplied to the interior chamber; determining, after the second period of time has elapsed, a second ratio between the amount of heat to be supplied to the interior chamber by the upper heating element and the amount of heat to be supplied to the interior chamber by the lower heating element; and causing heat to be supplied to the interior chamber by at least one of the upper and lower heating elements based on the determined second ratio.

In another aspect, temperature data characterizing a temperature of air within an upper region of an interior chamber of a cooking device above a cooking surface and configured to receive food to be cooked and a temperature of air within a lower region of the interior chamber below the cooking surface can be received. A ratio between an amount of heat to be supplied to the interior chamber by an upper heating element and an amount of heat to be supplied to the interior chamber by a lower heating element can be determined, and the upper heating element and the lower heating element can be disposed in the interior chamber at opposing surfaces of the interior chamber. Heat can be caused to be supplied to the interior chamber by at least one of the upper and lower heating elements based on the determined ratio.

One or more of the following features can be included in any feasible combination. For example, mode data characterizing a mode of cooking the food can be received, and the ratio can be determined based on the received mode data. For example, the causing of the heat to be supplied to the interior chamber can occur when the cooking device is operating in a pre-heating state. For example, a period of time beginning after the causing of the heat to be supplied to the interior chamber can be determined based on the determined ratio, and the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element can be adjusted after the determined period of time has elapsed. For example, the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element can be adjusted when the temperature of the air within the lower region of the interior chamber characterized by the temperature data is less than a target temperature. For example, second temperature data characterizing a second temperature of the air within the upper region of the interior chamber can be received, whether the second temperature is greater than or equal to the target temperature can be determined, and, the upper heating element and the lower heating element can be

deactivated in response to determining that the second temperature is greater than or equal to the target temperature.

In one aspect, a cooking device is described. The cooking device can include a housing defining an interior chamber, an upper heating element disposed within the interior chamber and on an upper surface of the housing and configured to supply heat to the interior chamber. The cooking device can also include a lower heating element disposed within the interior chamber, on a lower surface of the housing, proximate a cooking surface disposed within the interior chamber and configured to receive food to be cooked, and the lower heating element can be configured to supply heat to the interior chamber. The lower surface can be opposite the upper surface. The cooking device can also include a controller in operable communication with the upper heating element and the lower heating element. The controller can be configured to cause a first amount of heat to be generated by the upper heating element and no heat to be generated by the lower heating element when the controller is operating in a first mode, and, when a temperature of the air within an upper region of the interior chamber exceeds a predetermined threshold and when the controller is operating in the first mode, cause the controller to operate in a second mode in which a second amount of heat is generated by the upper heating element and a third amount of heat is generated by the lower heating element, the second and third amounts of heat determined by the controller and based on the temperature of the air within the upper region of the interior chamber and a temperature of a lower region of the interior chamber located below the cooking surface.

One or more of the following features can be included in any feasible combination. For example, the controller can be configured to receive data characterizing a mode of cooking the food and to determine the second and third amounts of heat based on the received data. For example, the cooking device can further include an upper temperature sensor disposed within the interior chamber and proximate the upper heating element, and the upper temperature sensor can be configured to measure the temperature of the air within the upper region of the interior chamber. For example, the cooking device can further include a lower temperature sensor disposed within the interior chamber and proximate the lower heating element and the cooking surface, and the lower temperature sensor can be configured to measure the temperature of at least one of the lower region of the interior chamber and the cooking surface. For example, the controller can be configured to cause the heat to be supplied to the interior chamber by at least one of the upper and lower heating elements when the controller is operating in a pre-heating mode. For example, the controller can be configured to adjust the second and third amounts of heat after a period of time, and the period of time can be determined based on the temperature of the air within the upper region of the interior chamber and the temperature of the air within the lower region of the interior chamber. For example, the controller can be configured to adjust the second and third amounts of heat when the temperature of the cooking surface is less than a target temperature. For example, the controller can be configured to deactivate the upper heating element when the temperature of the air within the interior chamber exceeds the predetermined threshold.

In another aspect, a system is provided and can include at least one data processor and memory storing instructions configured to cause the at least one data processor to perform operations described herein. The operations can include receiving temperature data characterizing a temperature of

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air within an upper region of an interior chamber of a cooking device, the upper region above a cooking surface, and characterizing a temperature of air within a lower region of the interior chamber, the lower region below the cooking surface; determining, based on the received temperature data, a first amount of heat to be supplied to the interior chamber by an upper heating element and a second amount of heat to be supplied to the interior chamber by a lower heating element, the upper heating element and the lower heating element being disposed in the interior chamber at opposing surfaces of the interior chamber; determining, during an operating mode in which a third amount of heat is supplied to the interior chamber by the upper heating element and no heat is supplied to the interior chamber by the lower heating element, whether the temperature of the air within the upper region of the interior chamber exceeds a predetermined threshold; and in response to determining that the temperature of the air within the upper region of the interior chamber exceeds the predetermined threshold, causing the first and second amounts of heat to be supplied to the interior chamber.

One or more of the following features can be included in any feasible combination. For example, the operations can further include receiving mode data characterizing a mode of cooking the food, and wherein the first and second amounts of heat are determined based on the received mode data. For example, the operations can further include adjusting the second and third amounts of heat after a period of time, and wherein the period of time is determined based on the temperature of the air within the upper region of the interior chamber and the temperature of the air within the lower region of the interior chamber. For example, the operations can further include adjusting the second and third amounts of heat when the temperature of the cooking surface is less than a target temperature. For example, the operations can further include causing the upper heating element and the lower heating element to deactivate in response to determining that the temperature of the air within the upper region of the interior chamber exceeds the predetermined threshold. For example, a portion of the temperature data can be received from a lower temperature sensor disposed within the interior chamber and proximate the lower heating element and the cooking surface, and the lower temperature sensor can be configured to measure the temperature of at least one of the lower region of the interior chamber and the cooking surface.

In another aspect, temperature data characterizing a temperature of air within an upper region of an interior chamber of a cooking device, the upper region above a cooking surface, and characterizing a temperature of air within a lower region of the interior chamber, the lower region below the cooking surface can be received. A first amount of heat to be supplied to the interior chamber by an upper heating element and a second amount of heat to be supplied to the interior chamber by a lower heating element, the upper heating element and the lower heating element being disposed in the interior chamber at opposing surfaces of the interior chamber can be determined based on the received temperature data, and the upper heating element and the lower heating element can be disposed in the interior chamber at opposing surfaces of the interior chamber. During an operating mode in which a third amount of heat is supplied to the interior chamber by the upper heating element and no heat is supplied to the interior chamber by the lower heating element, whether the temperature of the air within the upper region of the interior chamber exceeds a predetermined threshold can be determined. The first and second amounts

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of heat can be supplied to the interior chamber in response to determining that the temperature of the air within the upper region of the interior chamber exceeds the predetermined threshold.

One or more of the following features can be included in any feasible combination. For example, mode data characterizing a mode of cooking the food can be received, and the first and second amounts of heat can be determined based on the received mode data. For example, the second and third amounts of heat can be adjusted after a period of time, and the period of time can be determined based on the temperature of the air within the upper region of the interior chamber and the temperature of the air within the lower region of the interior chamber. For example, the second and third amounts of heat can be adjusted when the temperature of the cooking surface is less than a target temperature. For example, the upper heating element and the lower heating element can be deactivated in response to determining that the temperature of the air within the upper region of the interior chamber exceeds the predetermined threshold. For example, a portion of the temperature data can be received from a lower temperature sensor disposed within the interior chamber and proximate the lower heating element and the cooking surface, and the lower temperature sensor can be configured to measure the temperature of at least one of the lower region of the interior chamber and the cooking surface.

The details of one or more variations of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features and advantages of the subject matter described herein will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

These and other features will be more readily understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

FIG. 1A is a perspective view of a cooking device according to an embodiment;

FIG. 1B is a first perspective cross-sectional view of the cooking device of FIG. 1A;

FIG. 1C is a second perspective cross-sectional view of the cooking device of FIG. 1A;

FIG. 1D is a third perspective cross-sectional view of the cooking device of FIG. 1A;

FIG. 1E is a front side view of a user interface located on a front side of the cooking device of FIG. 1A;

FIG. 1F is a schematic diagram illustrating components of the cooking device of FIG. 1A;

FIG. 2 is a process flow diagram illustrating an exemplary process for initializing a pre-heat function of the cooking device of FIG. 1A;

FIG. 3A is a process flow diagram illustrating an exemplary process for pre-heating the cooking device of FIG. 1A;

FIG. 3B is a process flow diagram illustrating an exemplary process for determining a level of power to be applied to the heating elements of the cooking device of FIG. 1A; and

FIG. 4 is a process flow diagram illustrating an exemplary process for cooking a pizza with the cooking device of FIG. 1A.

It is noted that the drawings are not necessarily to scale. The drawings are intended to depict only typical aspects of the subject matter disclosed herein, and therefore should not be considered as limiting the scope of the disclosure.

DETAILED DESCRIPTION

Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the

structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings, and additional examples of specific system structures, functions, manufactures, uses, and related methods can be found in U.S. application Ser. Nos. 17/733,237, 17/663,582 and 18/079,781, each of which is incorporated by reference herein in its entirety. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present invention.

Further, in the present disclosure, like-named components of the embodiments generally have similar features, and thus within a particular embodiment each feature of each like-named component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape.

In general, cooking devices having at least one electric heating element and that are configured to cook various foods, including pizza, are provided. In some embodiments, the cooking device can include a housing having a base and a movable door coupled to the base that together define an interior cooking chamber. An upper heating element can be disposed within the interior chamber, and a lower heating element can be similarly disposed within the interior chamber opposite the upper heating element. A cooking surface, such as a cooking stone, can be disposed above the lower heating element within the cooking chamber such that food, such as a pizza, can be placed on top of the cooking stone when inserted into the interior chamber. The heating elements can be in operable communication with a controller configured to adjust the amount of heat supplied by the heating elements to the interior chamber to optimize the cooking of a food, such as pizza, placed inside the interior chamber.

FIGS. 1A-1F illustrate an exemplary cooking device **10** having at least one electric heating element according to an embodiment. The cooking device **10** can be used to cook food in a variety of cooking modes, including conductive and convective modes (e.g., Warm, Dehydrate, Smoker, Bake, Roast, Broil, Hi/Low, Sear, Pizza, etc.). The illustrated cooking device **10** includes a base **12** and a movable door **14** movably coupled to the base **12**, such as via a hinge (not shown). Together, the walls of the base **12** (e.g., a rear wall **12R**, a left wall **12L**, a right wall **12RI**, a top wall **12T**, and a bottom wall **12B** as shown in FIGS. 1A-1D), and the door **14** can be referred to generally as a housing and can define an interior cooking chamber **16** that is sized to receive a variety of food products and/or food containers (e.g., a baking tray, a rack, a cooking stone, etc.). The cooking device **10** can include a lower heating element **18A** disposed above a lower surface **12L** of the base **12** and an upper heating element **18B** that is disposed below an upper surface (not shown) of the base **12**. As referenced above, the base **12** can include a cooking surface **20**, such as a cooking stone

(also referred to as a pizza stone), upon which a food product (e.g., a pizza) can be placed during an operating procedure of the cooking device **10** and which can be disposed above the lower heating element **16A**. Each of the lower heating element **18A** and the upper heating element **18B** is configured to heat the cooking surface **20**, the interior cooking chamber **16** and/or the food product through conduction, convection, radiation, or a combination thereof. In some embodiments, the lower heating element **18A** can comprise a plurality of lower heating elements, and, in some embodiments, the upper heating element **18B** can comprise a plurality of upper heating elements. The cooking device **10** can also include a fan **22**, which can be operated to circulate air within the interior cooking chamber **16** during a variety of cooking modes. The fan **22** can be located within the interior cooking chamber **16** at a rear wall of the base **12**, as shown in FIGS. 1C and 1D. The cooking device **10** can also include a lower temperature sensor **24A** disposed below the cooking surface **20** and configured to measure a temperature of the air in a lower region **16A** of the interior cooking chamber **16** that is proximate a lower surface of the cooking surface **20** (not shown). The lower surface of the cooking surface **20** is disposed opposite an upper surface **20A** of the cooking surface **20** as shown in FIG. 1C, and the lower region **16A** can, in some embodiments, be bounded by the bottom wall **12B**, the left wall **12L**, the right wall **12RI**, the rear wall **12R**, the door **14**, and the lower surface of the cooking surface **20**. In some embodiments, the lower temperature sensor **24A** can be configured to measure a temperature of the lower surface of the cooking surface **20**. As such, in some embodiments, lower temperature data characterizing the lower surface of the cooking surface **20** and/or the air proximate the lower surface of the cooking surface **20** can be acquired and/or generated by the lower temperature sensor **24A** and provided to a controller (such as a controller **65**, described in further detail below) for use by the controller in determining the temperature of the air proximate the lower surface of the cooking surface **20** and/or the temperature of the lower surface of the cooking surface **20**. Similarly, the cooking device **10** can also include an upper temperature sensor **24B** disposed in the interior cooking chamber **16** and configured to measure a temperature of the air in an upper region **16B** of the interior cooking chamber **16** that is located above the cooking surface **20**. In some embodiments, the upper region **16B** can be bounded by the top wall **12T**, the left wall **12L**, the right wall **12RI**, the rear wall **12R**, the door **14**, and an upper surface **20B** of the cooking surface **20**. As such, in some embodiments, upper temperature data characterizing the air in the upper region **16B** of the interior cooking chamber **16** can be acquired and/or generated by the upper temperature sensor **24B** and provided to a controller (such as the controller **65** described below) for use by the controller in determining the temperature of the air proximate the lower surface of the cooking surface **20**. Exemplary lower and upper temperature sensors can include thermometers, thermocouples, NTC thermistors, and the like, or any collection/combination thereof. Lower and upper temperature sensors **24A**, **24B** are schematically illustrated in FIG. 1F, and their use is described in further detail below.

As explained above and as will be discussed in further detail below, the cooking device **10** can be pre-programmed to perform a unique cooking operation for cooking pizza based on a particular type or style of pizza, including Neapolitan style, thin-crust style, New York style, and pan style. An example of a Neapolitan style pizza can be a style of pizza featuring a relatively moist inner portion that includes pizza sauce (e.g., tomato-based sauce and the like)

and pizza toppings (e.g., cheese, meat, vegetables and the like) and that is surrounded by a crust that has been subjected to a high amount of heat to thereby cause “leoparding” of the crust (e.g., the formation of relatively small blisters of heavily-cooked regions of the crust and/or larger-sized bubbles of heavily-cooked regions of the crust). In some circumstances, it can be difficult to satisfactorily cook a Neapolitan style pizza because of the difference between an amount of heat required to sufficiently cook the inner portion of the pizza and the high amount of heat required to cause the leoparding of the crust described above. For example, by failing to adequately control the high amount of heat required for leoparding, or by applying the high amount of heat required for leoparding to the crust to the inner portion, the inner portion can become overcooked and thus unsuitable for eating.

The cooking device 10 can also be pre-programmed to perform a unique cooking operation for cooking a frozen pizza. In other aspects, the cooking device 10 can be configured to cook a pizza in accordance with user-defined custom cooking parameters. As such, in some embodiments, the cooking device 10 can include a user interface 50 that permits a user to configure the cooking device 10 for performing one or more pre-programmed pizza cooking modes corresponding to one or more of the above-described styles/types of pizza. As explained above, in some embodiments, these pre-programmed pizza cooking modes can include “Neapolitan” (for cooking Neapolitan style pizza), “Thin” (for cooking thin-crust style pizza), “NY” (for cooking New York style pizza), “Pan” (for cooking pan style pizza), “Frozen” (for cooking frozen pizza), and “Custom” (for cooking a pizza in accordance with user-defined custom cooking parameters). Any of the listed operations can be used in combination with one another, both in succession or at the same time. Further, these cooking modes can be combined with smoke generated by a smoke generation unit 26 (illustrated in FIGS. 1A, 1C, and 1D) integrated into the base 12. The structure and/or functionality of an exemplary embodiment of such a smoke generation unit is described in greater detail in U.S. application Ser. Nos. 17/733,237, 17/663,582 and 18/079,781 (incorporated by reference herein as described above). The user interface 50 can be located on an external portion of the cooking device 10, such as on a front face of the base 12, as seen, for example, in FIGS. 1A-1D. The user interface 50, which is shown in detail in FIG. 1E, can be in operable communication with a controller 65 (schematically illustrated in FIG. 1F). The user interface can be configured to receive inputs that cause the controller 65 to perform one or more actions responsive to the received inputs, as described in further detail below.

In some embodiments, the user interface 50 can include a variety of input buttons and controls that permit the user to configure the operation of the cooking device 10 by providing the above-described inputs. For example, in some embodiments, the user interface 50 can include a rotary function dial 51 which is configured to rotate when turned by a user, such that the user can select a pre-programmed cooking mode in which the cooking device 10 can operate. Exemplary pre-programmed cooking modes can include Warm, Dehydrate, Smoker, Bake, Roast, Broil, Hi/Low, Sear, and Pizza. The user interface 50 can also include a smoke generation button 52a which can cause the controller 65 to activate the smoke generation unit 26 described above and thereby supply smoke to the cooking chamber 16. The user interface 50 can illuminate a smoke generation icon 52b when the smoke generation unit 26 is activated. The user interface 50 can also include temperature and time buttons

53, 54, which, when pressed, can allow the user to adjust the cooking temperature and the cooking time of the cooking device 10, which are displayed above the buttons 53, 54 as illustrated in FIG. 1E. The user interface 50 can also include a start/stop button 55, which, when pressed by a user, causes the controller 65 to start and/or stop various operations of the cooking device 10 (e.g., pre-heating, cooking, etc.). The user interface 50 can also include a start/stop dial 56 which allows the user to perform additional functionality, such as scrolling through the pizza cooking modes referenced above and as described in further detail below. The user interface 50 can also include one or more indicators which indicate a status of the cooking device 10. For example, the user interface 50 can include a pizza cooking mode indicator 57, which can illuminate an indicator corresponding to one of the above-described pizza cooking modes when that mode is selected. Similarly, the user interface 50 can also include an “On” indicator 58 which illuminates when the cooking device is in an operational state.

As schematically illustrated in FIG. 1F and as referenced above, the cooking device 10 can include a controller 65 that is in operable communication with one or more of the components described above (e.g., the user interface 50, the lower temperature sensor 24A, the upper temperature sensor 24B, etc.). As shown, the controller 65 can include at least one processor 66 and a memory 67 storing instructions which, when executed by the processor 66, can cause the at least one data processor to perform one or more of the operations described elsewhere herein. The controller 65 can also include an input/output (I/O) interface 68 that enables the processor 66 to receive commands and/or data from other components of the cooking device 10 for use in performing the operations. For example, the controller 65 can receive, through the I/O interface 68, data characterizing temperature measurements made by the lower temperature sensor 24A and/or the upper temperature sensor 24B and provide that data to the processor 66 for use in performing operations requiring the received data as an input. Similarly, the controller 65 can receive, from the user interface 50 and via I/O interface 68, data characterizing inputs received from the user by the user interface 50 and provide that data to the processor 66 for use in performing operations that require the data received from the user interface 50 as input.

As shown in FIG. 1F, the cooking device 10 can also include a power supply 70 that is configured to supply power to the lower heating element 18A and the upper heating element 18B (in addition to other components of the cooking device 10 requiring power to operate). The power supply 70 can be in operable communication with the controller 65. As such, the power supply 70 can be configured to receive commands from the controller 65 (provided via the I/O interface 68) that cause the power supply 70 to provide electrical power to the lower heating element 18A and/or the upper heating element 18B to thereby cause the lower heating element 18A and/or the upper heating element 18B to activate and provide heat to the cooking chamber 16. In some embodiments, the power delivered to the lower heating element 18A and/or the upper heating element 18B can vary based on the commands received from the controller 65. For example, the level and/or duration of power supplied to the lower heating element 18A and/or the upper heating element 18B can vary over a given period of time based on power delivery instructions characterized by the commands received from the controller 65. As a result, the level and/or duration of heat being supplied to the cooking chamber 16 can vary over the given period of time based on the commands received from the controller 65.

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In some cooking scenarios, it can be advantageous to “pre-heat” the cooking device **10** prior to inserting the pizza into the interior cooking chamber **16**. FIG. **2** illustrates an exemplary process **200** for the initializing of a pre-heat function of the cooking device **10** and which can be implemented in some embodiments of the present subject matter. As shown, the process **200** begins at **201** when a user has selected the “Pizza” cooking mode with the function dial **51**. At **201**, the smoke generation icon **52b** on the user interface **50** is not illuminated (which indicates that the smoke generation unit **26** is not operational), and the user interface **50** displays the default cooking time. The process then proceeds to **202**, at which point the controller **65** monitors the position of the function dial **51** and determines whether the user is rotating the function dial **51**. In response to determining that the user is rotating the function dial **51**, the controller **65** then enters a function adjustment mode **203** in which the controller configures one or more components of the cooking device **10** in accordance with a selected one of the above-described cooking modes.

In response to determining that the user is not rotating the function dial **51**, the controller monitors the position of the start/stop dial **56** and determines, at **204**, whether the user is rotating the start/stop dial **56**. If the controller determines that the user is rotating the start/stop dial **58**, the controller then enters a pizza function adjustment mode **205**. When the controller enters the pizza function adjustment mode **205**, all available pizza cooking modes are illuminated on the user interface **50**. By rotating the start/stop dial **58**, the user can scroll through each of the pizza cooking modes (e.g., Neapolitan, Thin, NY, Pan, Frozen, and Custom) and thereby change the selected pizza cooking mode in accordance with their preference. In some embodiments, the controller **65** can also cause the user interface **50** to modify one or more aspects of the display based on the selected pizza cooking mode. For example, as the user rotates the start/stop dial **58** and scrolls through each of the pizza cooking modes, the controller **65** can update the time and/or temperature displayed on the user interface **50** to reflect a default length of cooking time and a default cooking temperature, respectively, that corresponds to the pizza cooking mode currently selected.

In response to determining, at **204**, that the user is not rotating the start/stop dial **58**, the controller **65** determines, at **206**, whether the user has pressed the time button **53**. If the controller determines that the user has pressed the time button **53**, the controller then enters a time adjustment mode **207** in which the user can interact with the user interface **50** to adjust the length of the cooking time from the default length of the cooking time associated with the selected pizza cooking mode to a desired length of the cooking time.

In response to determining, at **206**, that the user has not pressed the time button, the controller **65** determines, at **208**, whether the user has selected the custom function by rotating the start/stop dial **56** as discussed above. If the controller **65** determines that the user has selected the custom function, the controller then enters a custom adjustment mode **209**. When the controller is in the custom adjustment mode **209**, the user can adjust the cooking temperature and the length of the cooking time, and the user can set a single temperature value that serves as a set point for the temperature of the cooking chamber as well as a set point for the temperature of the bottom of the cooking surface **20**. When the user adjusts the cooking temperature such that the set point for the temperature of the bottom of the cooking surface **20** exceeds a first predetermined temperature threshold, the

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controller can determine an optimized bottom set point temperature based on the cooking temperature selected by the user.

In response to determining, at **208**, that the user has not selected the custom function, the controller **65** determines, at **210**, whether the user has pressed a probe button located on the user interface **50** (not shown). If the controller **65** determines that the user has pressed a probe button, the controller, at **211**, causes an error notification to be presented to the user.

In response to determining, at **210**, that the user has not pressed the probe button, the controller **65** determines, at **212**, whether the user has pressed the smoke generation button **52a**. If the controller **65** determines that the user has pressed the smoke generation button **52a**, the controller then causes a smoke operating mode to activate **213**. The controller **65** can determine, at **214**, whether the user has pressed the smoke generation button **52a** a second time and cause, at **215**, the smoke generation unit **26** to exit the smoke operating mode in response to the user pressing the smoke generation button **52a** a second time.

After making the determinations described above at **212** and **214**, the controller **65** can determine, at **216**, whether the user has pressed the start/stop button **55**. If the controller **65** determines that the user has not pressed the start/stop button **55**, the process then returns to **202**, and the controller **65** monitors the position of the function dial **51** to determine whether the user has rotated the function dial **51** as described above.

If the controller **65** determines that the user has pressed the start/stop button **55**, the controller **65** determines, at **217** whether the configuration of the cooking device **10** (e.g. the previously-determined settings/functions) is the same as that which was active during a previous period of time (e.g., the immediately preceding five minutes, etc.). If so, the pre-heat function is skipped and the initialization process **200** ends at **218**.

If the controller **65** determines that the configuration of the cooking device **10** is not the same as that which was active during the previous period of time, the controller **65** determines at **219** whether the smoke mode has been activated. If so, the controller **65** causes the smoke generation unit **26** to initiate a fuel ignition process **220**, an example of which is described in U.S. patent application Ser. No. 17/663,582, hereby incorporated by reference herein in its entirety, and, once complete, enables the cooking device **10** for pre-heating as described below with respect to FIG. **3A**, and the initialization process ends at **222**. If not, the controller **65** enables the cooking device **10** for pre-heating as described below with respect to FIG. **3A** and the initialization process ends at **223**. In some embodiments, if the fuel ignition process **220** has previously occurred, the pre-heating function is skipped.

FIG. **3A** illustrates an exemplary process **300A** for controlling the pre-heating of the cooking device **10** and which can be implemented in some embodiments of the present subject matter. The process **300** begins at **301**, wherein the initialization process **200** as described above with respect to FIG. **2** has ended at either of **222** or **223**.

At **301**, the controller **65** has begun to pre-heat the cooking chamber **16** and has updated the user interface **50** to indicate that the cooking device **10** is pre-heating the cooking chamber **16** in accordance with the pizza function selected by the user as described above. At **302**, the controller **65** determines the amount of power that the power supply **70** should provide the lower heating element **18A** and/or the upper heating element **18B** and subsequently

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provides instructions to the power supply 70 that cause the power supply 70 to provide the determined amounts of power to be supplied to each of the lower and upper heating elements 18A, 18B (and thereby cause the pre-heating of the cooking chamber 16). The process by which the controller 65 determines the amount of power that the power supply 70 should provide to the lower heating element 18A and to the upper heating element 18B is described in detail below with respect to FIG. 3B.

While the cooking device is undergoing pre-heating, the controller 65 determines, at 303, whether the pre-heating of the cooking chamber 16 has occurred for a first predetermined amount of time (e.g., 35 minutes, etc.). In response to determining that the pre-heating of the cooking chamber 16 has occurred for the first predetermined amount of time, the controller 65 enables the cooking device 10 for the cooking of pizza as described below with respect to FIG. 4, and the pre-heating process 300 ends at 304. In response to determining that the pre-heating of the cooking chamber 16 has not occurred for the first predetermined amount of time, the controller 65 determines, at 305, whether the pre-heating of the cooking chamber 16 has occurred for a second predetermined amount of time (e.g., 25 minutes, etc.) that is less than the first predetermined amount of time. If the controller 65 determines that the pre-heating of the cooking chamber 16 has occurred for the second predetermined amount of time, the controller 65 determines, at 306, whether the cooking temperature is less than or equal to a second predetermined temperature threshold. In some embodiments, the second predetermined temperature threshold may be the same as the first determined temperature threshold described above. If the controller 65 determines, as part of the power amount determinations performed at 302, that the pre-heating process is complete, the controller 65 can enable the cooking device 10 for the cooking of pizzas as described below with respect to FIG. 4 and end the pre-heating process at 307.

As explained above, at 302 of the pre-heating control process 300A, the controller 65 can determine the amount of power that should be provided to each of the lower heating element 18A and the upper heating element 18B by the power supply 70. FIG. 3B illustrates an exemplary power output calculation process 300B to be performed at 302 and which can be implemented in some embodiments of the present subject matter. As shown, at the beginning of the process 302, the controller 65 determines, at 308, whether the temperature measured with the upper temperature sensor 24B is greater than or equal to a target temperature. If the controller 65 determines that the temperature measured with the upper temperature sensor 24B is greater than or equal to the target temperature, the controller 65 then determines, at 309, whether the temperature measured with the lower temperature sensor 24A is greater than or equal to the target temperature. If the controller 65 determines that temperature measured with the lower temperature sensor 24A is less than the target temperature, the controller 65 then, at 310, causes the power supply 70 to provide no power to the upper heating element 18B and to provide full power to be supplied to the lower heating element 18A. If the controller 65 determines, at 308, that the temperature measured with the upper temperature sensor 24B is less than the target temperature, the controller then determines, at 311, whether the temperature measured with the lower temperature sensor 24A is greater than or equal to the target temperature. If the controller 65 determines at 311 that the temperature measured with the lower temperature sensor 24A is greater than or equal to the target temperature, the controller 65

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then, at 312, causes the power supply 70 to provide no power to the lower heating element 18A and full power to be supplied to the upper heating element 18B.

If the controller 65 determines at 311 that the temperature measured with the lower temperature sensor 24A is less than the target temperature, the controller 65 then, at 313, determines a power ratio characterizing the relationship between the amount of power to be supplied to the upper heating element 18B relative to the amount of power to be supplied to the lower heating element 18A. In some embodiments, the power ratio can characterize a discrepancy between an actual temperature, measured by a temperature sensor (such as the lower temperature sensor 24A and/or the upper temperature sensor 24B), and a target temperature. To determine the power ratio, in some embodiments and as explained above, the controller 65 can receive, from the upper temperature sensor 24B, upper temperature data characterizing the temperature of the air within the cooking chamber 16, and the controller 65 can receive, from the lower temperature sensor 24A, lower temperature data characterizing the temperature of the bottom of the cooking surface 20. The controller 65 can determine the power ratio based on the received upper temperature data, the received lower temperature data, and the target temperature.

The controller 65 also, at 313, determines a lower heating element energy value, which characterizes a portion of available energy that is dedicated to the lower heating element 18A and that is above and beyond a predetermined minimum lower heating element power value when the temperature measured with the lower temperature sensor 24A and the temperature measured with the upper temperature sensor 24B are below their respective target temperatures. Additionally, the controller 65, at 313, determines an upper heating element energy value, which characterizes a portion of available energy that is dedicated to the upper heating element 18B and that is above and beyond a predetermined minimum upper heating element power value when the temperature measured with the lower temperature sensor 24A and the temperature measured with the upper temperature sensor 24B are below their respective target temperatures. The controller 65 determines the lower heating element energy value based on the determined power ratio, the minimum lower heating element power value, and the minimum upper heating element power value. Additionally, the controller 65 determines the upper heating element energy value based on the minimum lower heating element power value, the minimum upper heating element power value, and the determined lower heating element energy value.

The controller 65, at 313, determines a lower heating portion value that characterizes a portion of the available energy allocated to the lower heating element 18A when each of the temperature measured with the lower temperature sensor 24A and the temperature measured with the upper temperature sensor 24B are below their respective target temperatures. And, similarly, the controller 65, at 313, determines an upper heating portion value that characterizes a portion of the available energy allocated to the upper heating element 18B when each of the temperature measured with the lower temperature sensor 24A and the temperature measured with the upper temperature sensor 24B are below their respective target temperature. The controller 65 determines the lower heating portion value based on the minimum lower heating element power value and the determined lower heating element energy value, and the controller 65 can determine the upper heating portion value based

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on the minimum upper heating element power value and the determined upper heating element energy value.

In some embodiments, the controller 65 can determine the lower heating element energy value without regard to the predetermined minimum lower heating element power value. In this case, the lower heating element energy value characterizes a portion of available energy that is dedicated to the lower heating element 18A. As such, the controller 65 can determine the lower heating element energy value based on the determined power ratio. And, as such, the controller 65 can determine the lower heating portion value based on the determined lower heating element energy value. Similarly, in some embodiments, the controller 65 can, at 313, determine the upper heating element energy value without regard to the predetermined minimum upper heating element power value. In this case, the upper heating element energy value characterizes a portion of available energy that is dedicated to the upper heating element 18B. As such, the controller 65 can determine the upper heating element energy value based on the determined lower heating element energy value. And, as such, the controller 65 can determine the upper heating portion value based on the determined upper heating element energy value.

Once the controller 65 has made the above-described determinations at 313, the controller 65, at 314, causes the lower heating element 18A to operate at full power and the upper heating element 18B to be deactivated. At 315, the controller 65 determines whether the temperature measured with the lower temperature sensor 24A is greater than or equal to the target temperature. If the controller 65 determines that the temperature measured with the lower temperature sensor 24A is less than the target temperature, the controller 65 then determines, at 316 and based on the lower heating portion value, a portion of a predefined time interval during which the lower heating element 18A is activated and whether that portion of the predefined time interval has elapsed. If so, the controller 65, at 317, causes the lower heating element 18A to deactivate and the upper heating element 18B to activate at full power. If not, the controller 65 re-determines, at 315, whether the temperature measured with the lower temperature sensor 24A is greater than or equal to the target temperature. If, at 315, the controller 65 determines that the temperature measured with the lower temperature sensor 24A is greater than or equal to the target temperature, the controller bypass the determination at 316 and causes the lower heating element 18A to deactivate and the upper heating element 18B to activate at full power at 317.

At 318, the controller 65 receives data characterizing temperatures measured by the upper temperature sensor 24B and determines, based on the received temperature data, whether the temperature of the air measured with the upper temperature sensor 24B is greater than or equal to the target temperature. If the controller 65 determines at 318 that the temperature measured with the upper temperature sensor 24B is less than the target temperature, the controller 65 then determines, at 319 and based on the upper heating portion value, a portion of a predefined time interval during which the upper heating element 18B is activated and whether that portion of the predefined time interval has elapsed. If so, the controller 65 then returns to 308 to re-determine whether the temperature measured with the upper temperature sensor 24B is greater than or equal to the target temperature and performs one or more of the operations based on the results of that determination as described elsewhere herein. If not, the controller 65 then returns to 318 to re-determine whether the temperature measured with the upper temperature sensor

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24B is greater than or equal to the target temperature. If the controller 65 determines, at 318, that the temperature measured with the upper temperature sensor 24B is greater than or equal to the target temperature, the controller 65 deactivates the upper heating element 18B and the lower heating element 18A at 320. At 321, the controller 65 determines whether the cooking device 10 is operating in a pre-heat state. If so, the system ends the pre-heating process at 304 (as also illustrated in FIG. 3A). If not, the controller 65, at 322, updates the user interface 50 to display the default temperature setting for the selected pizza function as well as a countdown timer that indicates the amount of time remaining during the pre-heating process.

As explained above, the controller 65, at 309, determines whether the temperature measured with the lower temperature sensor 24A is greater than or equal to the target temperature. If the controller 65 determines that the temperature measured with the lower temperature sensor 24A is greater than or equal to the target temperature, the controller 65 then performs the functionality described above at 321.

Additionally, as explained above, if, at 309, the controller 65 determines that the temperature measured with the lower temperature sensor 24A is less than the target temperature, the controller 65 then, at 310, causes no power to be supplied to the upper heating element 18B and full power to be supplied to the lower heating element 18A. When this occurs, the controller 65 determines, at 323, whether the lower heating element 18A has been activated for a first predefined period of time (e.g., 30 seconds). If the controller 65 determines that the lower heating element 18A has been activated for the first predefined period of time, the controller 65 returns to 308 to re-determine whether the temperature measured with the upper temperature sensor 24B is greater than or equal to the target temperature. If the controller 65 determines that the lower heating element 18A has been activated for less than the first predefined period of time, the controller 65 then, at 324, determines whether the temperature measured with the lower temperature sensor 24A is greater than or equal to the target temperature. If so, the controller 65, at 325, causes the lower and upper heating elements 18A, 18B to deactivate and then performs the functionality described above at 321.

In addition, as explained above, if the controller 65 determines, at 311, that the temperature measured with the lower temperature sensor 24A is greater than or equal to the target temperature, the controller 65 then, at 312, causes no power to be supplied to the lower heating element 18A and full power to be supplied to the upper heating element 18B. When this occurs, the controller 65 determines, at 326, whether the upper heating element 18B has been activated for a second predefined period of time (e.g., 30 seconds). If the controller 65 determines that the upper heating element 18B has been activated for a second predefined period of time, the controller 65 returns to 308 to re-determine whether the temperature measured with the upper temperature sensor 24B is greater than or equal to the target temperature and performs one or more of the operations based on the results of that determination as described elsewhere herein. If the controller 65 determines that the lower heating element 18A has been activated for less than the second predefined period of time, the controller 65 then, at 327, determines whether the temperature measured with the upper temperature sensor 24B is greater than or equal to the target temperature. If so, the controller 65, at 328, turns off the upper and lower heating elements 18A, 18B and then performs the functionality described above at 321.

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However, in some embodiments, the pre-heat function can be skipped in response to the occurrence of one or more events. For example, in some embodiments, the pre-heat function can be skipped by a user pushing and holding down the start/stop dial **56** for a certain period of time. And, as another example, in some embodiments, the pre-heat function can be skipped if the temperature of the air within the cooking chamber exceeds a target temperature for a particular pizza cooking mode at the time the pizza cooking mode is selected as described above.

FIG. 4 illustrates an exemplary process **400** for controlling the pizza cooking function of the cooking device **10** and which can be implemented in some embodiments of the present subject matter. In some embodiments, process **400** can occur after the pre-heating process **300a** described above has concluded. The process **400** begins at **401**, at which point the controller **65** causes power to be supplied as appropriate to the upper heating element **18B** and the lower heating element **18A** and such that the pre-heat set temperature is maintained in accordance with the pizza cooking mode selected in the manner described above. The controller **65** then, at **402**, determines whether the user has pressed the start/stop button **55** on the user interface **50**. If not, the controller **65** then, at **403**, determines whether a certain period of time (e.g., an hour, etc.) has elapsed since the beginning of process **400** without the user having pressed the start/stop button **55**. If not, the user interface displays a periodic notification to the user. If so, the controller **65** causes the cooking device **10** to deactivate, ending the process at **404**.

In response to determining at **402** that the user has pressed the start/stop button **55**, then, the controller **65** then activates one or more of the heating elements to begin to cook the pizza at **405**. The controller **65** causes the upper and/or lower heating elements **18A**, **18B** to activate in accordance with the pizza function selected by the user. For example, if the user has selected Neapolitan as the pizza cooking mode, the controller **65** only activates the upper heating element **18B** and deactivates the lower heating element **18A**. For example, if the user has selected one of the other pizza cooking modes described above, the controller **65** determines the amount of power that the power supply **70** is to provide to the upper heating element **18B** and to the lower heating element **18A** independently by executing the process **300B** described above with respect to FIG. 3B. Once the controller **65** has made that determination, the controller **65** then causes the power supply **70** to supply the determined amount of power to each of the upper heating element and the lower heating element **18A**.

If the user has selected the Neapolitan pizza function, the controller **65** determines, at **406**, whether the temperature measured with the upper temperature sensor **24B** exceeds a first predetermined threshold (e.g., 750 degrees Fahrenheit). If so, then the controller **65** causes the upper heating element **18B** to deactivate and the lower heating element **18A** to activate at full power at **407**. After adjusting the operation of the heating elements at **407** and when in the Neapolitan mode, the controller **65** determines, at **408**, whether the temperature measured with the upper temperature sensor **24B**, as measured by the upper temperature sensor **24B**, exceeds a second predetermined threshold (e.g., 700 degrees Fahrenheit). If not, then, the controller **65** determines, at **409**, whether the temperature measured with the lower temperature sensor **24A** exceeds a predetermined threshold (e.g., 700 degrees Fahrenheit). If the controller **65** determines that the temperature measured with the lower temperature sensor **24A** has exceeded the predetermined thresh-

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old, (e.g., 700 degrees Fahrenheit), then the controller **65**, at **410**, deactivates the lower heating element **18A** and the upper heating element **18B** (if not previously deactivated), determines whether a predetermined period of time (e.g., 5 seconds) has passed, and returns to **408** to re-determine whether the temperature of the air in the cooking chamber **16** exceeds the first predetermined threshold (e.g., 700 degrees Fahrenheit). If not, the controller returns to **408** to monitor the temperature of the air in the cooking chamber **16** and re-determine whether the temperature measured with the upper temperature sensor **24B** exceeds the first predetermined threshold (e.g., 700 degrees Fahrenheit). In some embodiments, the controller can reactivate the lower heating element **18A** and/or the lower heating element **18B** in response to determining whether the temperature measured with the lower temperature sensor **24A** and/or the temperature measured with the upper temperature sensor **24B** has exceeded a predetermined threshold. In some embodiments, if the controller **65** determines that the temperature measured with the upper temperature sensor **24B** has exceeded the predetermined threshold, (e.g., 700 degrees Fahrenheit), then the controller **65** deactivates the lower heating element **18A** and the upper heating element **18B** (if not previously deactivated) and re-executes the process **300B** described above to pre-heat the cooking chamber **16** as quickly as possible.

During the period of time that the cooking device **10** is cooking the pizza, a user can press the start/stop button **55** on the user interface **50** to pause cooking. As such, during cooking, if the controller **65** determines, at **412**, that the start/stop button **55** has been pressed by the user, the controller **65** causes, at **413**, the heating elements **18A** and/or **18B** to deactivate and a notification to be displayed on the user interface **50** indicating that the cooking process has ended and instructing the user to add additional food to the cooking chamber **16** and to resume cooking by pressing the start/stop button **55**. When this occurs, the controller **65**, at **414**, re-executes the process **300B** described above to maintain the level of heat required for the selected pizza cooking mode in the cooking chamber **16**.

If the controller **65** determines at **412** that the user has not pressed the stop/start button **55** when the pizza is being cooked, cooking continues in accordance with the time and temperature settings of the selected pizza cooking mode. As the cooking device **10** continues to cook the pizza, the controller **65** monitors the length of time during which the pizza is being cooked. When the controller **65** determines, at **415**, that the time allotted for cooking of the pizza and specified by the selected pizza cooking mode has elapsed, the controller **65**, at **416**, causes a notification to be displayed on the user interface indicating that the pizza has been cooked and instructing the user to remove the cooked pizza. Once this occurs, the controller **65**, at **414**, re-executes the process **300B** described above to pre-heat the cooking chamber **16** as quickly as possible following the cooking of the first pizza. After initializing the second pre-heat process at **414**, the controller **65** monitors the temperature measured with the lower temperature sensor **24A** and the temperature measured with the upper temperature sensor **24B** and determines whether those temperatures have reached their respective target temperatures at **417**. In response to determining, at **417**, that both target temperatures have been met, the controller **65** causes a notification to appear on the user interface **50** indicating that the second pre-heat process has ended at **419** and that the cooking device **10** is ready to end the pre-heating process described above and cause the cooking process **400** to begin once again at **401** in response

to determining that a user has the start/stop button **55** on the user interface **50** at **420**. If the controller **65** determines, at **420**, that a user has failed to press the start/stop button to restart the cooking process, and, at **421**, that a predefined period of time (e.g., an hour) has elapsed since the end of the second pre-heating process, and, at **22**, that no other cooking process is actively running, the controller **65** causes the cooking device to turn off at **423**.

One or more aspects or features of the subject matter described herein can be realized in digital electronic circuitry, integrated circuitry, specially designed application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs) computer hardware, firmware, software, and/or combinations thereof. These various aspects or features can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which can be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device. The programmable system or computing system may include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

These computer programs, which can also be referred to as programs, software, software applications, applications, components, or code, include machine instructions for a programmable processor, and can be implemented in a high-level procedural language, an object-oriented programming language, a functional programming language, a logical programming language, and/or in assembly/machine language. As used herein, the term “machine-readable medium” refers to any computer program product, apparatus and/or device, such as for example magnetic discs, optical disks, memory, and Programmable Logic Devices (PLDs), used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor. The machine-readable medium can store such machine instructions non-transitorily, such as for example as would a non-transient solid-state memory or a magnetic hard drive or any equivalent storage medium. The machine-readable medium can alternatively or additionally store such machine instructions in a transient manner, such as for example as would a processor cache or other random access memory associated with one or more physical processor cores.

To provide for interaction with a user, one or more aspects or features of the subject matter described herein can be implemented on a computer having a display device, such as for example a cathode ray tube (CRT) or a liquid crystal display (LCD) or a light emitting diode (LED) monitor for displaying information to the user and a keyboard and a pointing device, such as for example a mouse or a trackball, by which the user may provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well. For example, feedback provided to the user can be any form of sensory feedback, such as for example visual feedback, auditory feedback, or tactile feedback; and input from the user may be received in any form, including, but not limited to, acoustic, speech, or tactile input. Other possible input devices include, but are not limited to, touch

screens or other touch-sensitive devices such as single or multi-point resistive or capacitive trackpads, voice recognition hardware and software, optical scanners, optical pointers, digital image capture devices and associated interpretation software, and the like.

Certain exemplary implementations have been described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the systems, devices, and methods disclosed herein. One or more examples of these implementations have been illustrated in the accompanying drawings. Those skilled in the art will understand that the systems, devices, and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary implementations and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary implementation may be combined with the features of other implementations. Such modifications and variations are intended to be included within the scope of the present invention. Further, in the present disclosure, like-named components of the implementations generally have similar features, and thus within a particular implementation each feature of each like-named component is not necessarily fully elaborated upon.

Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise.

One skilled in the art will appreciate further features and advantages of the invention based on the above-described implementations. Accordingly, the present application is not to be limited by what has been particularly shown and described, except as indicated by the appended claims. All publications and references cited herein are expressly incorporated by reference in their entirety.

What is claimed is:

1. A method comprising:

receiving first temperature data characterizing a temperature of air within an upper region of an interior chamber of a cooking device, the upper region being above a cooking surface, and second temperature data characterizing a temperature of air within a lower region of the interior chamber, the lower region being below the cooking surface;

determining, based on the received first and second temperature data, a ratio between an amount of heat to be supplied to the interior chamber by an upper heater and an amount of heat to be supplied to the interior chamber by a lower heater, the upper and lower heaters being disposed in the interior chamber at opposing surfaces of the interior chamber;

causing heat to be supplied to the interior chamber by at least one of the upper and lower heaters based on the determined ratio; and

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adjusting an amount of heat supplied by at least one of the upper and lower heaters after a period of time, wherein the period of time is determined based on the determined ratio.

2. The method of claim 1, further comprising receiving mode data characterizing a mode of cooking food, wherein the mode data includes a target temperature, and wherein the ratio is further determined based on the received mode data.

3. The method of claim 1, the causing of the heat to be supplied to the interior chamber occurs when the cooking device is operating in a pre-heating state.

4. The method of claim 3, further comprising adjusting the amount of the heat supplied to the interior chamber by the upper heater and the amount of the heat supplied to the interior chamber by the lower heater when the temperature of the air within the lower region of the interior chamber is less than a target temperature.

5. The method of claim 4, further comprising: receiving third temperature data characterizing a subsequent temperature of the air within the upper region of the interior chamber;

determining whether the subsequent temperature is greater than or equal to the target temperature; and in response to determining that the subsequent temperature is greater than or equal to the target temperature, causing the upper heater and the lower heater to deactivate.

6. The method of claim 2, wherein the method is carried out by a cooking system having a user interface, the method further comprising receiving, via the user interface, a user input characterizing a selection of the mode of cooking the food from a plurality of cooking modes.

7. The method of claim 6, wherein the plurality of cooking modes comprise one or more predefined modes for cooking pizza and a custom mode configured for cooking a pizza in accordance with user-defined custom cooking parameters.

8. The method of claim 7, wherein the period of time is further determined based on the mode data.

9. The method of claim 8, further comprising: comparing the temperature of air within the lower region to the target temperature;

determining whether the period of time has elapsed; responsive to determining that the temperature of the air within the lower region is less than the target temperature and that the period of time has elapsed, causing the lower heater to deactivate and the upper heater to activate at a maximum power; and

responsive to determining that the temperature of the air within the lower region is greater than or equal to the target temperature, causing the lower heater to deactivate and the upper heater to activate at the maximum power.

10. The method of claim 1, further comprising: determining a lower heater energy value based on the determined ratio, the lower heater energy value characterizing a portion of available energy allocated to the lower heater and which is higher than a predetermined minimum lower heater power value;

determining a lower heating portion value based on the lower heater energy value, the lower heating portion value characterizing a portion of available energy allocated to the lower heater when a first temperature measured with a lower temperature sensor is below a first predetermined threshold temperature and a second temperature measured with an upper temperature sensor is below a second predetermined threshold temperature; and

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wherein the period of time is determined based on the determined lower heating portion value.

11. A cooking device comprising:

an interior chamber having a cooking surface;
a first temperature sensor configured acquire first temperature data characterizing a temperature of air within an upper region of the interior chamber, the upper region being above the cooking surface;

a second temperature sensor configured acquire second temperature data characterizing a temperature of air within a lower region of the interior chamber, the lower region being below the cooking surface;

a first heater configured to provide heat to the upper region;

a second heater configured to provide heat to the lower region; and

a controller communicatively coupled to the first and second temperature sensors and the first and second heaters, the controller configured to perform operations comprising

receiving first temperature data characterizing a temperature of air within the upper region from the first temperature sensor and second temperature data characterizing a temperature of air within the lower region from the second temperature sensor,

determining, based on the received first and second temperature data, a ratio between an amount of heat to be supplied to the interior chamber,

causing heat to be supplied to the interior chamber by at least one of the first and second heaters based on the determined ratio, and

adjusting an amount of heat supplied by at least one of the first and second heaters after a period of time, wherein the period of time is determined based on the determined ratio.

12. The cooking device of claim 11, wherein the controller is configured to perform operations further comprising receiving mode data characterizing a mode of cooking food, wherein the mode data includes a target temperature, and wherein the ratio is further determined based on the received mode data.

13. The cooking device of claim 11, wherein the causing of the heat to be supplied to the interior chamber occurs when the cooking device is operating in a pre-heating state.

14. The cooking device of claim 13, wherein the controller is configured to perform operations further comprising adjusting the amount of the heat supplied to the interior chamber by the first heater and the amount of the heat supplied to the interior chamber by the second heater when the temperature of the air within the lower region of the interior chamber is less than a target temperature.

15. The cooking device of claim 14, wherein the controller is configured to perform operations further comprising: receiving third temperature data characterizing a subsequent temperature of the air within the upper region of the interior chamber from the first temperature sensor; determining whether the subsequent temperature is greater than or equal to the target temperature; and in response to determining that the subsequent temperature is greater than or equal to the target temperature, causing the first heater and the second heater to deactivate.

16. The cooking device of claim 11, further comprising a user interface, wherein the controller is configured to receive, via the user interface, a user input characterizing a selection of the mode of cooking the food from a plurality of cooking modes.

17. The cooking device of claim 16, wherein the plurality of cooking modes comprise one or more predefined modes for cooking pizza and a custom mode configured for cooking a pizza in accordance with user-defined custom cooking parameters.

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18. The cooking device of claim 17, wherein the period of time is further determined based on the mode data.

19. The cooking device of claim 18, wherein the controller is configured to perform operations further comprising:
comparing the temperature of air within the lower region 10
to the target temperature;
determining whether the period of time has elapsed;
responsive to determining that the temperature of the air
within the lower region is less than the target tempera-
ture and that the period of time has elapsed, causing the 15
lower heater to deactivate and the upper heater to
activate at a maximum power; and
responsive to determining that the temperature of the air
within the lower region is greater than or equal to the
target temperature, causing the lower heater to deacti- 20
vate and the upper heater to activate at maximum
power.

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